

A Project Report on

**MULTIMODAL PARKINSON'S DISEASE
DETECTION USING CROSS-MODAL
ATTENTION FUSION OF HANDWRITING
AND SPEECH BIOMARKERS**

Submitted in partial fulfillment for the award of the degree of

**Bachelor of Technology
in
Computer Science and Engineering**

by

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Amaravati, Andhra Pradesh

April 2026

DECLARATION

We hereby declare that the project entitled "MULTIMODAL PARKINSON'S DISEASE DETECTION USING CROSS-MODAL ATTENTION FUSION OF HANDWRITING AND SPEECH BIOMARKERS" submitted by us, for the award of the degree of Bachelor of Technology in Computer Science and Engineering at VIT-AP University is a record of bonafide work carried out by us under the supervision of our project guide.

We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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CERTIFICATE

This is to certify that the Senior Design Project titled "MULTIMODAL PARKINSON'S DISEASE DETECTION USING CROSS-MODAL ATTENTION FUSION OF HANDWRITING AND SPEECH BIOMARKERS" submitted by Tanguturi Venkata Thanuj (22BCE20003), Katam Krishna Chaitanya (22BCE7359), and Pathakuntla Narendar Reddy (22BCE7707) is in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, and is a record of bonafide work done under our guidance. The contents of this project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for award of any degree or diploma and the same is certified.

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ABSTRACT

Parkinson's Disease (PD) is the second most prevalent neurodegenerative disorder worldwide, affecting over 10 million individuals and imposing a significant clinical and economic burden. Conventional diagnosis primarily relies on subjective clinical assessments such as the Unified Parkinson's Disease Rating Scale (UPDRS) and costly neuroimaging techniques like DaTSCAN, which typically range from \$1,500 to \$4,000 per scan. These approaches are often limited to advanced disease stages and are not readily accessible in resource-constrained settings, highlighting the need for **scalable, non-invasive, and cost-effective early screening solutions**.

This work presents a **multimodal deep learning framework** for Parkinson's Disease detection that leverages **handwriting and speech biomarkers**, two mechanistically complementary modalities that can be acquired using consumer-grade devices. The proposed system integrates three synergistic modules. First, a **handwriting analysis module** combines 16 clinically motivated spatial biomarkers with an EfficientNet-B0 architecture enhanced by a Convolutional Block Attention Module (CBAM), achieving **93.6% accuracy with an AUC of 0.985**. Second, a **speech analysis module** employs a multi-pathway architecture incorporating self-supervised XLS-R 300M representations, Mel-spectrogram CNNs, MFCC features, and acoustic descriptors fused via attention mechanisms, achieving **91.7% accuracy with an AUC of 0.971**. Third, a novel **Cross-Modal Attention Fusion Network (CMAFN)** is introduced, which utilizes **bidirectional Transformer-based cross-attention** and a **Gated Multimodal Unit (GMU)** to model interactions between handwriting and speech representations, achieving **96.9% accuracy and an AUC-ROC of 0.999**, thereby surpassing existing multimodal approaches.

To ensure rigorous evaluation, **patient-level stratified 5-fold cross-validation** is employed, eliminating data leakage and ensuring realistic generalization. The system is implemented as a set of **interactive Flask-based web applications**, enabling real-time inference in under 3 seconds on CPU-only hardware, making it suitable for deployment in **resource-limited and telemedicine environments**. Experimental results demonstrate that **explicit cross-modal interaction learning between handwriting micrographia and speech dysarthria yields significant synergistic improvements** over unimodal approaches.

Keywords

Parkinson's Disease, Multimodal Deep Learning, Cross-Modal Attention, EfficientNet, XLS-R, Handwriting Biomarkers, Speech Analysis, Gated Multimodal Unit, Transfer Learning, CBAM.

ACKNOWLEDGEMENT

It is our pleasure to express with deep sense of gratitude to our project guide for the constant guidance, continual encouragement, and understanding throughout the development of this project. The association has not been confined to academics only; it has been a great opportunity to work with an intellectual expert in the field of deep learning and biomedical signal processing.

We would like to express our gratitude to the Chancellor, Vice-Chancellors, and the Dean of the School of Computer Science and Engineering, VIT-AP University, for providing an excellent environment to pursue this research and for the inspiration during the course of this program.

We express our heartfelt thanks to all teaching staff and members of VIT-AP University for their timely encouragement and guidance. We would also like to thank the developers of the NewHandPD dataset and the Italian Parkinson's Voice and Speech dataset (IEEE DataPort) for making clinical data publicly available for research.

Finally, we express our gratitude to our parents and friends who supported and encouraged us throughout this journey.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
PD	Parkinson's Disease
HC	Healthy Control
CMAFN	Cross-Modal Attention Fusion Network
CBAM	Convolutional Block Attention Module
GMU	Gated Multimodal Unit
XLS-R	Cross-Lingual Speech Representations
MFCC	Mel-Frequency Cepstral Coefficients
CNN	Convolutional Neural Network
DNN	Deep Neural Network
MLP	Multi-Layer Perceptron
UPDRS	Unified Parkinson's Disease Rating Scale
DaTSCAN	Dopamine Transporter Scan
HNRR	Harmonics-to-Noise Ratio
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
FFT	Fast Fourier Transform
TTA	Test-Time Augmentation
CV	Cross-Validation
SVM	Support Vector Machine
SMOTE	Synthetic Minority Oversampling Technique
PCA	Principal Component Analysis
SPP	Spatial Pyramid Pooling
BN	Batch Normalization
GELU	Gaussian Error Linear Unit
SE	Squeeze-and-Excitation
FFN	Feed-Forward Network
MC	Monte Carlo
API	Application Programming Interface
Flask	Python Web Framework
GPU	Graphics Processing Unit

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Parkinson's Disease (PD) is the second most prevalent neurodegenerative disorder worldwide, primarily affecting the elderly population and characterized by the progressive degeneration of dopaminergic neurons in the **substantia nigra pars compacta**. This neuronal loss leads to a deficiency of dopamine, a critical neurotransmitter responsible for regulating motor control, coordination, and smooth execution of voluntary movements. As the disease progresses, patients exhibit cardinal motor symptoms such as **resting tremor, rigidity, bradykinesia (slowness of movement), and postural instability**. However, clinical manifestation of these symptoms typically occurs only after approximately **60–80% of dopaminergic neurons have already been irreversibly damaged**, significantly limiting the effectiveness of therapeutic interventions initiated at later stages.

Globally, Parkinson's Disease affects **over 10 million individuals**, with projections indicating a **doubling of prevalence by 2040** due to aging populations and increased life expectancy. The socioeconomic burden is substantial, with annual healthcare and associated costs exceeding **\$52 billion in the United States alone**. Current diagnostic approaches rely heavily on **subjective clinical rating scales**, such as the Unified Parkinson's Disease Rating Scale (UPDRS), and **expensive neuroimaging techniques** like DaTSCAN and PET scans. These methods are not only costly and resource-intensive but also lack sensitivity in detecting early-stage PD, especially in under-resourced or remote settings where access to neurologists is limited.

In recent years, there has been growing interest in identifying **non-invasive, objective, and scalable biomarkers** for early PD detection. Among these, **handwriting and speech signals** have emerged as highly promising modalities due to their strong physiological correlation with neuromotor control mechanisms.

Handwriting impairments, commonly referred to as **micrographia**, manifest as a progressive reduction in writing size, irregular stroke patterns, increased variability in stroke width, and decreased motor smoothness. These changes reflect **fine motor control degradation** and can be quantitatively analyzed using spatial and geometric features such as curvature, entropy, fractal dimension, and stroke variability.

Similarly, speech impairments in PD, including **dysarthria and hypophonia**, are characterized by reduced vocal intensity, monotonic pitch, articulatory imprecision, and abnormal voice quality measures such as jitter and shimmer. These alterations arise from dysfunction in the **phonatory and respiratory subsystems**, and can be effectively captured using acoustic and spectral features derived from short audio recordings.

Importantly, handwriting and speech represent **complementary biomarkers**:

- Handwriting reflects **peripheral motor control** and fine motor coordination.
- Speech reflects **central neuromotor and phonatory control**.

This complementary nature makes them ideal for **multimodal analysis**, where integrating both modalities can significantly enhance diagnostic accuracy compared to single-modality approaches.

Advancements in **deep learning and multimodal fusion techniques** have enabled the development of automated systems capable of extracting complex patterns from these data sources. In particular, **cross-modal attention mechanisms and Transformer-based architectures** allow models to learn interdependencies between modalities, enabling more robust and accurate predictions. Furthermore, the availability of **consumer-grade devices** such as smartphones, tablets, and microphones makes data acquisition simple, scalable, and accessible, facilitating large-scale screening.

Therefore, leveraging handwriting and speech biomarkers within an **AI-driven multimodal framework** presents a powerful, cost-effective, and non-invasive solution for early Parkinson’s Disease detection, with the potential to significantly improve clinical outcomes and reduce healthcare burden.

1.2 PROBLEM STATEMENT

Conventional diagnosis of Parkinson’s Disease (PD) primarily relies on clinical assessment using the **Unified Parkinson’s Disease Rating Scale (UPDRS)** and advanced neuroimaging techniques such as **DaTSCAN (SPECT)** or **PET scans**, which typically cost between **\$1,500 and \$4,000 per scan**. While these approaches remain the clinical standard, they present several inherent limitations. The UPDRS is **subjective in nature**, relying on clinician expertise and observational judgment, which introduces inter-rater variability and reduces reliability, particularly in early-stage detection. Neuroimaging methods, although more objective, are **expensive, resource-intensive, and not widely accessible**, especially in rural or under-resourced healthcare environments. Furthermore, both methods are generally effective only when **significant dopaminergic neuron loss (60–80%) has already occurred**, thereby limiting their usefulness for early diagnosis.

A critical **diagnostic gap** exists between the onset of early, subtle symptoms and formal clinical diagnosis. During this window, patients often exhibit mild manifestations such as **micrographia (changes in handwriting)** and **speech impairments (dysarthria and hypophonia)**, which are frequently overlooked or not systematically quantified in routine clinical practice. This delay in diagnosis results in missed opportunities for **early therapeutic intervention**, which is known to improve patient outcomes, slow disease progression, and enhance quality of life.

In parallel, existing computational approaches for PD detection often suffer from several challenges:

- **Single-modality dependence**, where models rely only on either handwriting or speech, limiting diagnostic robustness.
- **Lack of multimodal interaction modeling**, where simple fusion techniques fail to capture the complementary nature of motor and speech biomarkers.
- **Overfitting and data leakage**, particularly in medical datasets, where improper validation strategies (e.g., sample-level splits instead of patient-level splits) lead to inflated performance metrics.
- **High computational requirements**, with many deep learning models requiring GPU-based inference, making them unsuitable for real-world deployment in low-resource settings.

Given these limitations, there is a strong need for a **scalable, objective, and non-invasive diagnostic framework** that can:

1. Utilize **readily available data sources** such as handwriting samples and short speech recordings collected via consumer-grade devices (smartphones, microphones, tablets).
2. Extract both **engineered clinical biomarkers** and **deep learned representations** to capture subtle neuromotor impairments.
3. Effectively **fuse multiple modalities** by learning cross-modal relationships rather than relying on simple concatenation or late fusion techniques.
4. Ensure **robust and unbiased evaluation** through patient-level stratified validation to prevent data leakage.
5. Achieve **high diagnostic accuracy and reliability**, comparable to or exceeding existing unimodal approaches.
6. Remain **computationally efficient and deployable**, enabling real-time inference (<3 seconds) on CPU-only systems without requiring specialized hardware.

Therefore, the central research question addressed in this project is:

Can non-invasive, low-cost biomarkers derived from handwriting and speech—captured using consumer hardware—be effectively integrated within a deep learning framework, specifically through cross-modal attention mechanisms, to achieve clinically meaningful Parkinson’s Disease detection accuracy while ensuring scalability, robustness, and real-world deployability in resource-constrained environments?

Addressing this challenge requires not only designing accurate models but also ensuring **practical usability**, **generalization across patients**, and **integration into deployable systems**, thereby bridging the gap between research innovation and real-world clinical application.

1.3 OBJECTIVES

The primary objective of this work is to develop a **robust, scalable, and non-invasive multimodal deep learning framework** for early detection of Parkinson’s Disease (PD) by leveraging complementary biomarkers derived from handwriting and speech. The specific objectives of this study are as follows:

1. Handwriting Biomarker Extraction and Analysis

To design and implement an advanced **feature extraction pipeline** that derives **16 clinically relevant spatial biomarkers** from spiral and wave handwriting samples using **OpenCV-based morphological image processing techniques**. These features include stroke width statistics, contour roughness, curvature measures, entropy, fractal dimension, and Hu moments, which collectively capture **fine motor impairments such as tremor, micrographia, and reduced motor control**. Additionally, to integrate these engineered features with deep learning representations obtained from **EfficientNet-based convolutional neural networks enhanced with attention mechanisms (CBAM)**, enabling a hybrid analysis of both handcrafted and learned features.

2. Multi-Pathway Speech Analysis Model

To develop a **comprehensive multi-pathway speech processing framework** that captures diverse aspects of vocal impairment associated with Parkinson's Disease. This includes:

- Leveraging **self-supervised speech representations (XLS-R 300M)** to capture high-level phonetic and prosodic information.
- Utilizing **Mel-spectrogram-based convolutional neural networks** to learn spectral-temporal patterns.
- Extracting **MFCC features with temporal derivatives** to represent timbral and resonance characteristics.
- Incorporating **Praat-based acoustic features** such as jitter, shimmer, and harmonics-to-noise ratio (HNR) to quantify voice quality abnormalities.

These heterogeneous features are to be fused using **multi-head cross-attention mechanisms** and further enhanced through a **hybrid deep learning and classical machine learning stacking approach**, ensuring robust and generalized speech-based PD detection.

3. Design of Cross-Modal Attention Fusion Network (CMAFN)

To propose and implement a novel **Cross-Modal Attention Fusion Network (CMAFN)** that effectively integrates handwriting and speech modalities. The model aims to:

- Learn **bidirectional Transformer-based cross-attention**, where handwriting features attend to speech features and vice versa.
- Capture **interdependencies between motor and vocal biomarkers**, enabling richer feature representations.
- Incorporate a **Gated Multimodal Unit (GMU)** to dynamically balance the contribution of each modality, preventing dominance of one modality over the other.
- Improve overall diagnostic performance by exploiting the **complementary nature of multimodal data**.

4. Robust Validation and Generalization

To ensure **reliable and unbiased model evaluation**, all modules are validated using **patient-level stratified 5-fold cross-validation**, which prevents **data leakage** by ensuring that samples from the same patient do not appear in both training and testing sets. Additionally, to incorporate **regularization strategies** such as modality dropout, label smoothing, and early stopping to enhance generalization and reduce overfitting, thereby ensuring clinical reliability of the proposed system.

5. Real-Time Deployment and Practical Usability

To develop and deploy the entire system as **interactive Flask-based web applications**, integrating handwriting and speech modules into a unified interface. The deployment aims to:

- Enable **real-time inference (< 3 seconds)** on standard CPU hardware.
- Eliminate the need for **specialized GPU infrastructure**, ensuring accessibility in resource-constrained environments.
- Support **user-friendly input methods**, including drawing interfaces and audio uploads.
- Provide **interpretable outputs** such as risk levels (LOW, MODERATE, HIGH) and confidence estimates for clinical decision support.

6. Achieving Clinically Meaningful Performance

To achieve **high diagnostic accuracy and reliability**, targeting performance metrics such as **accuracy above 95% and AUC-ROC close to 1.0**, thereby demonstrating the effectiveness of multimodal fusion over unimodal approaches and establishing the system as a **practical tool for early Parkinson's Disease screening**.

Summary of Objectives

In essence, this work aims to bridge the gap between **clinical need and technological capability** by developing a **multimodal, attention-driven, and deployable AI system** that enables **early, accurate, and accessible detection of Parkinson's Disease** using everyday devices.

1.4 SCOPE OF THE PROJECT

This project focuses on the development of a binary classification system for Parkinson's Disease (PD), aimed at distinguishing Parkinson's patients (PD) from Healthy Controls (HC) using two complementary, non-invasive modalities: handwriting drawings (spiral and wave patterns) and short speech recordings. The system is designed as an early screening tool rather than a definitive diagnostic solution, assisting in the identification of individuals who may require further clinical evaluation.

The proposed framework leverages multimodal deep learning techniques, specifically combining handwriting and speech biomarkers through a Cross-Modal Attention Fusion Network (CMAFN), to improve diagnostic accuracy. The system is intended to operate using consumer-grade hardware, such as smartphones, tablets, and microphones, thereby ensuring accessibility, scalability, and usability in real-world environments, including rural and resource-constrained settings.

Scope of Functional Capabilities

Within the defined scope, the system includes the following capabilities:

- **Handwriting Analysis:** Processing spiral and wave drawings to extract 16 clinically relevant spatial biomarkers along with deep visual features using convolutional neural networks.
- **Speech Analysis:** Processing short-duration audio recordings (< 8 seconds) to extract acoustic, spectral, and self-supervised features using a multi-pathway architecture.
- **Multimodal Fusion:** Integrating handwriting and speech features using bidirectional Transformer-based cross-attention and a Gated Multimodal Unit (GMU) to capture complementary information.
- **Risk Prediction Output:** Providing a binary classification result (PD vs HC) along with interpretable risk levels (e.g., LOW, MODERATE, HIGH) and confidence scores.
- **Real-Time Deployment:** Supporting interactive web-based usage via Flask, with CPU-only inference (< 3 seconds) and no dependency on GPU hardware.

Dataset Scope and Experimental Constraints

The experimental evaluation of this project is limited to the following datasets:

- **Handwriting Dataset (NewHandPD / Combined Dataset):**
 - Total samples: 3,264 images
 - Subjects: 65 individuals (including PD and healthy controls)
 - Data type: Spiral and wave drawings
 - Balanced class distribution for robust training
- **Speech Dataset (Italian Parkinson's Voice and Speech – IEEE DataPort):**
 - Total recordings: 831 audio samples
 - Subjects: 61 individuals
 - Includes multiple speech tasks (e.g., sustained vowels, reading, free speech)

- Standardized to 16 kHz and ≤ 8 seconds duration

To ensure valid and unbiased evaluation, all experiments are conducted using patient-level stratified 5-fold cross-validation, preventing overlap of patient data across training and testing sets.

Deployment Scope

The system is implemented as a modular Flask-based web application, integrating:

- Independent handwriting and speech analysis modules
- A unified multimodal fusion interface
- User-friendly input mechanisms (drawing canvas and audio upload)

The deployed model is optimized using quantization techniques (~85 MB model size) and designed for low-latency inference, making it suitable for deployment in clinical support systems, telemedicine platforms, and mobile health applications.

Limitations of the Current Scope

Despite its strengths, the current implementation has the following limitations:

- **Binary Classification Only:** The system distinguishes only between PD and HC, without providing disease severity staging (e.g., UPDRS scoring or Hoehn & Yahr stages).
- **Limited Dataset Diversity:** The model is trained and evaluated on specific datasets, which may not fully capture global population diversity.
- **Language Dependency in Speech:** The speech model is primarily trained on the Italian Parkinson's dataset, limiting direct generalization to other languages without retraining.
- **Short-Duration Inputs:** The system operates on short audio samples (< 8 seconds) and static handwriting images, without incorporating continuous or longitudinal monitoring.

Future Scope

The following extensions are identified as potential future work:

- **Severity Prediction:** Extending the model to predict disease progression and severity scores (e.g., UPDRS regression).

- Longitudinal Monitoring: Incorporating time-series data to track disease progression over time.
- Multi-Language and Cross-Dataset Generalization: Expanding the speech model to support multiple languages and diverse datasets.
- Integration with Additional Modalities: Including other biomarkers such as gait analysis, facial expressions, or wearable sensor data.
- Mobile Application Deployment: Developing lightweight mobile applications for large-scale screening and real-time monitoring.

Summary of Scope

In summary, this project is scoped to deliver a highly accurate, multimodal, and deployable screening system for Parkinson's Disease using handwriting and speech biomarkers, with a focus on early detection, accessibility, and real-world applicability, while acknowledging limitations that provide clear directions for future research and development.

CHAPTER 2

RELATED WORK

2.1 HANDWRITING-BASED DETECTION

The analysis of handwriting as a clinical biomarker for Parkinson’s Disease (PD) originates from early neurological observations of **micrographia**, a hallmark symptom characterized by a progressive reduction in writing size, speed, and stroke consistency due to impaired dopaminergic control. These abnormalities reflect underlying deficits in **fine motor coordination, tremor regulation, and neuromuscular control**, making handwriting a highly informative and non-invasive modality for early PD screening. Initial computational approaches focused on **kinematic analysis** using digitising tablets, where dynamic features such as pen velocity, acceleration, pressure, and stroke timing were captured. Zham et al. (2017) demonstrated the effectiveness of this approach by extracting such features and applying Support Vector Machines (SVM), achieving an accuracy of 87.4%. However, despite strong performance, these methods required specialized hardware, limiting scalability and real-world deployment.

The introduction of publicly available datasets such as **NewHandPD** by Pereira et al. (2018) marked a significant shift toward **image-based handwriting analysis**, enabling researchers to utilize standard scanned or photographed drawings instead of digitiser-based inputs. Their work established a baseline using **AlexNet fine-tuning**, achieving 85.0% accuracy on spiral and meander drawings. Building on this foundation, Kumar et al. (2023) improved performance to 87.2% by leveraging **ResNet50 with ImageNet pre-training**, highlighting the benefits of transfer learning in medical imaging tasks with limited data. Zhang et al. (2024) further explored **EfficientNet-B3 architectures combined with aggressive data augmentation techniques** such as mixup, cutmix, and random transformations to enhance generalization. Liu et al. (2022) investigated **ensemble learning strategies**, combining multiple deep architectures including ResNet, VGG, and Inception through majority voting, demonstrating improved robustness over single-model approaches. Additionally, Gupta and Mishra (2023) introduced **GAN-based data augmentation using StyleGAN2 with DenseNet121**, addressing the challenge of limited dataset size by generating synthetic handwriting samples that improved model performance.

A key advancement in this domain was achieved by Li et al. (2024), who introduced **spatial attention mechanisms within convolutional neural networks**, enabling the model to selectively focus on **tremor-affected stroke regions and irregular contours** in handwriting images. This approach achieved a state-of-the-art accuracy of 90.1%, demonstrating the importance of attention-based feature learning in capturing clinically relevant patterns. Despite these advancements, most existing methods rely either on **pure deep learning approaches or handcrafted features independently**, without fully leveraging the complementary strengths of both. To address these limitations, the present work adopts a **hybrid dual-pathway approach**, combining **engineered spatial biomarkers with deep convolutional representations**. Specifically, 16 clinically motivated features—such as stroke width variability, contour roughness, curvature statistics, entropy, fractal dimension, and Hu moments—are extracted using OpenCV-based image processing, capturing

interpretable motor impairment indicators. In parallel, an **EfficientNet-B0 model enhanced with Convolutional Block Attention Module (CBAM)** is used to learn high-level visual representations from raw handwriting images. These two pathways are integrated using a **stacking-based meta-learning framework (logistic regression)**, which effectively combines handcrafted and learned features to improve predictive performance. This hybrid approach achieves an accuracy of approximately **93.6% with an AUC of 0.985**, surpassing the prior state-of-the-art by **+3.47 percentage points**, while maintaining strong generalization through **patient-level stratified cross-validation**.

Overall, handwriting-based detection has evolved from hardware-dependent kinematic analysis to **scalable, image-based deep learning systems**, and further to **hybrid multimodal feature frameworks**, demonstrating its potential as a **practical, non-invasive, and cost-effective biomarker** for early Parkinson's Disease screening.

2.2 SPEECH-BASED DETECTION

Speech biomarker research for Parkinson's Disease (PD) has evolved significantly over the past two decades, progressing from traditional acoustic feature analysis to advanced deep learning and self-supervised representation learning. Speech impairments in PD, commonly referred to as **dysarthria and hypophonia**, arise due to neuromuscular dysfunction affecting the respiratory, phonatory, and articulatory subsystems. These impairments manifest as reduced vocal intensity, monotonic pitch, imprecise articulation, and irregular voice quality, making speech a highly informative and non-invasive modality for early PD detection. Early foundational work by Little et al. (2009) demonstrated that acoustic features such as **jitter (frequency variation)**, **shimmer (amplitude variation)**, and **harmonics-to-noise ratio (HNR)**—extracted using tools like Praat—are highly sensitive indicators of Parkinsonian speech. This work established **MFCC (Mel-Frequency Cepstral Coefficients) combined with Support Vector Machines (SVM)** as a strong baseline, achieving approximately 86% accuracy and forming the foundation for subsequent research in speech-based PD detection.

Building upon this, Tsanas et al. (2024) significantly expanded the feature space by introducing a comprehensive set of **132 acoustic features**, including nonlinear and statistical descriptors of voice signals, and demonstrated that ensemble methods such as **XGBoost** can achieve improved performance of around 86.7%. These studies highlighted the importance of **feature engineering** in capturing subtle voice impairments associated with PD. However, reliance on handcrafted features alone limits the model's ability to learn complex temporal and spectral patterns present in speech signals.

With the advancement of deep learning, researchers began exploring neural network-based approaches to automatically learn representations from raw or minimally processed audio data. Wang et al. (2024) demonstrated the effectiveness of **1D Convolutional Neural Networks (1D-CNNs)** applied to raw MFCC sequences, achieving an accuracy of 83.6%. Sakar et al. (2022) further improved temporal modeling by combining **CNN and BiLSTM architectures**, capturing both spatial and sequential dependencies in speech signals, and reported an accuracy of 84.2%. Ferrario et al. (2023) explored **Transformer-based architectures**, leveraging self-attention mechanisms to model long-range dependencies in speech, achieving 85.3% accuracy. While these approaches improved representation learning, they were still

limited by the availability of labeled data and the complexity of training deep models from scratch.

A major breakthrough in speech-based PD detection came with the introduction of **self-supervised learning models**, particularly **Wav2Vec 2.0**, which learn rich speech representations from large-scale unlabeled audio data. Orobic et al. (2024) demonstrated that fine-tuning Wav2Vec 2.0 on PD datasets can achieve 86.8% accuracy even with limited labeled samples, significantly outperforming traditional supervised approaches. These models capture **high-level phonetic, prosodic, and articulation features**, making them highly effective for medical speech analysis. Despite these advancements, most existing approaches rely on either **single-feature pipelines or single-model architectures**, limiting their ability to fully capture the multi-dimensional nature of speech impairments in PD. To address these limitations, the present work proposes a **multi-pathway hybrid framework** that integrates both deep learning and classical machine learning approaches. Specifically, the system combines four complementary feature pathways: (1) **XLS-R 300M self-supervised embeddings**, trained on 436,000 hours of multilingual speech across 128 languages, capturing rich phonetic and contextual information; (2) **Mel-spectrogram-based CNN features**, capturing spectral-temporal energy distributions; (3) **MFCC features with temporal derivatives**, representing timbral and resonance characteristics; and (4) **Praat-based acoustic measures**, including jitter, shimmer, HNR, and other clinically validated voice quality indicators. These heterogeneous features are fused using **multi-head cross-attention mechanisms**, enabling the model to learn interactions across different feature spaces.

In addition, a **hybrid stacking framework** is employed, combining predictions from deep learning models and classical machine learning classifiers (such as SVM, Random Forest, and Gradient Boosting) using a **Ridge-regularized logistic regression meta-learner**, thereby improving robustness and generalization. The system incorporates advanced regularization techniques, including PCA-based dimensionality reduction, dropout, label smoothing, and early stopping, to prevent overfitting. Importantly, all evaluations are conducted using **patient-level stratified 5-fold cross-validation**, ensuring that no speaker data leaks between training and testing sets, which is critical for reliable performance estimation in medical applications.

As a result, the proposed speech module achieves an accuracy of approximately **91.7% with an AUC-ROC of 0.971**, significantly surpassing prior state-of-the-art methods by over **2.2 percentage points**, while maintaining strong generalization and clinical relevance. This demonstrates that combining **self-supervised representations with multi-pathway feature fusion and hybrid learning strategies** provides a powerful and scalable solution for speech-based Parkinson’s Disease detection.

2.3 MULTIMODAL FUSION APPROACHES

Multimodal Parkinson’s Disease (PD) detection has emerged as a promising research direction in recent years, driven by the understanding that different physiological signals capture **complementary aspects of neuromotor dysfunction**. While unimodal approaches based on handwriting or speech alone have shown strong performance, they are inherently limited in capturing the full spectrum of motor and non-motor impairments associated with PD. Multimodal systems aim to overcome these limitations by integrating multiple data sources, thereby improving robustness, generalization, and diagnostic accuracy. However, this field remains relatively

nascent, with ongoing research exploring various fusion strategies, modality combinations, and architectural designs.

Early multimodal approaches primarily relied on **late fusion techniques**, where independent models were trained on each modality and their outputs were combined at the decision level. For instance, Zhang et al. (2023) combined **gait accelerometer signals and speech features** using a CNN+LSTM framework with late fusion, achieving an accuracy of 91.2%. While effective, such approaches do not explicitly model interactions between modalities, thereby limiting their ability to capture cross-modal dependencies. To address this, researchers explored **early fusion techniques**, where raw or intermediate features from different modalities are concatenated and processed jointly. Ren et al. (2022) investigated **handwriting and MRI fusion** using a combination of ResNet for image processing and 3D-CNN for volumetric brain imaging, achieving 89.4% accuracy. Although early fusion allows joint feature learning, it often struggles with **heterogeneous data representations** and modality imbalance.

More recent works have shifted toward **attention-based fusion mechanisms**, which dynamically weigh the importance of different modalities. Quan et al. (2024) proposed a system combining **speech and facial expression analysis** using cross-attention mechanisms, achieving 88.6% accuracy. These approaches highlight the importance of **learning inter-modal relationships**, rather than treating modalities as independent sources of information. However, many of these models employ **unidirectional or shallow attention mechanisms**, which may not fully capture the bidirectional dependencies between modalities.

A significant advancement in multimodal PD detection was achieved by Vasquez-Correa et al. (2023), who proposed a **hierarchical fusion network** integrating three modalities—gait, speech, and handwriting—achieving a state-of-the-art accuracy of 94.3%. This work demonstrated that combining multiple modalities can substantially improve performance compared to unimodal systems. However, the inclusion of additional modalities such as gait often requires **specialized sensors or wearable devices**, limiting scalability and real-world deployment.

Despite these advancements, several challenges remain in multimodal PD detection, including: (1) **ineffective modeling of cross-modal interactions**, where simple concatenation or late fusion fails to capture complex relationships; (2) **increased system complexity and data acquisition requirements** when incorporating multiple modalities; and (3) **lack of robust evaluation strategies**, with some studies not employing patient-level validation, leading to potential data leakage and overestimated performance.

To address these limitations, the present work introduces a novel **Cross-Modal Attention Fusion Network (CMAFN)** that leverages **bidirectional Transformer-based cross-attention** to explicitly model interactions between handwriting and speech modalities. Unlike prior approaches, the proposed method allows **handwriting features to attend to speech features and vice versa**, enabling the model to learn which aspects of motor impairment correlate with specific vocal abnormalities. This bidirectional design captures deeper interdependencies between modalities, leading to more informative and discriminative feature representations. Additionally, a **Gated Multimodal Unit (GMU)** is employed to dynamically balance the contributions of each modality, preventing dominance by any single modality and improving robustness in cases where one modality may be noisy or missing.

Another key advantage of the proposed approach is its focus on **practical deployability**. By utilizing only handwriting and speech—both of which can be

collected using consumer-grade devices—the system avoids reliance on expensive or specialized hardware such as MRI scanners or wearable gait sensors. Furthermore, techniques such as **modality dropout** are incorporated during training to ensure that the model remains robust even when one modality is unavailable at inference time. As a result, the proposed CMAFN achieves an accuracy of **96.94% with an AUC-ROC of 0.9995**, surpassing the previous best multimodal system (94.3%) by **+2.64 percentage points**, despite using only two modalities. This demonstrates that **architectural sophistication, particularly in modeling cross-modal interactions, can outperform simply increasing the number of modalities**. The results highlight the effectiveness of combining **attention-based fusion with complementary biomarkers**, establishing the proposed framework as a **state-of-the-art, scalable, and clinically relevant solution** for multimodal Parkinson’s Disease detection.

Study	Modality	Method	Accuracy	AUC
Pereira et al. (2018)	Handwriting	CNN (AlexNet)	85.0%	—
Zham et al. (2017)	Handwriting	Kinematic+SVM	87.4%	—
Li et al. (2024)	Handwriting	Spatial Attention CNN	90.1%	—
Sakar et al. (2022)	Speech	SVM + MFCC	86.0%	0.91
Orobic et al. (2024)	Speech	Wav2Vec 2.0	86.8%	0.93
Mughal et al. (2023)	Speech	CapsNet + MFCC	84.5%	—
Vasquez-Correa et al. (2023)	Multi (3)	Hierarchical Fusion	94.3%	—
This Work: Handwriting	Handwriting	EffNet-B0+CBAM+Stack	93.6%	0.985
This Work: Speech	Speech	XLS-R+CrossAttn+Stack	91.7%	0.971
This Work: CMAFN	Multi (2)	X-Modal Transformer+GMU	96.9%	0.999

Table 2.1: Comparative Survey of PD Detection Literature

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 OVERALL SYSTEM DESIGN

The proposed system is designed as a **modular, hierarchical, and end-to-end multimodal framework** for Parkinson’s Disease (PD) detection, comprising three tightly integrated yet independently deployable modules: the **Handwriting Analysis Module**, the **Speech Analysis Module**, and the **Cross-Modal Attention Fusion Network (CMAFN)**. The overall architecture follows a **three-stage pipeline**, where unimodal feature extraction and prediction are first performed independently, followed by a higher-level fusion stage that integrates complementary information across modalities. This design ensures **flexibility, scalability, and robustness**, allowing each module to function as a standalone diagnostic component or as part of a unified multimodal system. Each module is implemented as a **Flask-based web application**, enabling real-time interaction, seamless integration, and deployment on standard CPU hardware without requiring GPU acceleration.

At the input level, the system accepts two types of non-invasive data: **handwriting images** (spiral or wave drawings resized to standardized resolutions) and **speech recordings** (audio signals sampled at 16 kHz and limited to short durations for efficient processing). These inputs are preprocessed through dedicated pipelines tailored to each modality, ensuring normalization, noise reduction, and compatibility with downstream models. The architecture is specifically designed to support **graceful degradation**, meaning that the system can still operate effectively even if only one modality is available, ensuring robustness in real-world scenarios where complete data may not always be present.

The first module, the **Handwriting Analysis Module**, employs a **dual-pathway ensemble architecture** that combines both handcrafted and deep learned features. One pathway consists of a **Residual Multi-Layer Perceptron (MLP)** that processes 16 engineered spatial biomarkers extracted using OpenCV-based morphological operations, capturing clinically interpretable indicators such as stroke variability, curvature, entropy, and fractal complexity. The second pathway utilizes an **EfficientNet-B0 convolutional neural network enhanced with Convolutional Block Attention Module (CBAM)**, which learns high-level visual representations from raw handwriting images while focusing on salient regions affected by tremor and motor irregularities. The outputs of these two pathways are integrated using a **stacking-based meta-learner (logistic regression)**, resulting in a robust and accurate handwriting-based prediction.

The second module, the **Speech Analysis Module**, is designed as a **multi-pathway feature extraction and fusion framework** that captures diverse aspects of speech impairment in PD. It incorporates four complementary pathways: (1) **XLS-R 300M self-supervised embeddings**, which provide rich phonetic and contextual representations learned from large-scale multilingual data; (2) **Mel-spectrogram-based convolutional neural networks**, which capture spectral-temporal patterns in speech signals; (3) **MFCC features with temporal derivatives**, representing timbral and resonance characteristics; and (4) **Praat-based acoustic features**, including jitter, shimmer, and harmonics-to-noise ratio, which are clinically validated indicators of voice quality degradation. These heterogeneous feature representations are projected

into a common embedding space and fused using **multi-head cross-attention mechanisms**, enabling the model to learn interdependencies across feature types. Additionally, a **hybrid deep learning and classical machine learning ensemble**, combined through a stacking meta-learner, enhances robustness and generalization. The third and most critical component, the **Cross-Modal Attention Fusion Network (CMAFN)**, serves as the core integration layer of the system. This module takes the learned embeddings from the handwriting and speech encoders and projects them into a shared latent space. It then applies **bidirectional Transformer-based cross-attention**, allowing handwriting features to attend to speech features and vice versa, thereby capturing complex relationships between motor and vocal impairments. To further refine the fusion process, a **Gated Multimodal Unit (GMU)** is employed, which dynamically controls the contribution of each modality through a learnable gating mechanism, ensuring balanced and context-aware integration. The fused representation is then passed through fully connected layers to produce the final classification output.

The system is optimized for **real-time performance and practical deployment**, with all modules designed to operate efficiently on CPU-only environments, achieving inference times of less than 3 seconds. Model optimization techniques such as **quantization and efficient architecture design** are employed to reduce memory footprint and computational overhead. Furthermore, the architecture incorporates **robust training strategies**, including patient-level stratified cross-validation, modality dropout, and regularization techniques, to ensure reliable performance and generalization across diverse patient populations.

Overall, the proposed system design emphasizes **modularity, interpretability, efficiency, and scalability**, enabling it to function both as a research framework and as a practical tool for **accessible, non-invasive, and real-time Parkinson’s Disease screening** in clinical and remote settings.

3.2 MODULE INTERACTIONS

The proposed system is structured to enable **seamless interaction between unimodal and multimodal components**, where the handwriting and speech modules serve dual and complementary roles within the overall architecture. Firstly, each module functions as an **independent diagnostic engine**, capable of performing unimodal Parkinson’s Disease (PD) detection using either handwriting or speech data alone. This ensures flexibility in deployment scenarios where only a single modality may be available. Secondly, both modules act as **pre-trained feature encoders**, generating high-quality, modality-specific embeddings that are subsequently utilized by the **Cross-Modal Attention Fusion Network (CMAFN)** for multimodal integration. This dual-role design enhances both **modularity and reusability**, allowing each component to be trained, evaluated, and deployed independently while still contributing to the overall system performance when combined.

The interaction between modules is facilitated through a **hierarchical feature flow**, where raw inputs are first processed within their respective modules to produce both **classification outputs and intermediate feature representations**. These learned embeddings capture rich modality-specific information—such as motor control patterns from handwriting and phonatory characteristics from speech—which are then projected into a shared latent space within the CMAFN. The fusion module leverages these representations to model **cross-modal dependencies**, enabling more accurate and context-aware predictions.

To ensure robustness in real-world conditions, the system incorporates a **modality dropout mechanism** during training, where each modality (handwriting or speech) is randomly masked with a probability of **25%**. This encourages the CMAFN to learn meaningful representations even in the absence of one modality, thereby enabling **graceful degradation** during inference. In scenarios where only one modality is available, the system can still produce reliable predictions by relying on the available input and learned fallback representations, making it highly suitable for practical deployment in diverse environments.

Each module outputs a **probabilistic prediction for binary classification (PD vs. HC)**, along with an associated **confidence estimate**. To enhance the reliability of these predictions, the system employs **Monte Carlo (MC) Dropout during inference**, which performs multiple stochastic forward passes to estimate prediction uncertainty. This results in a **calibrated uncertainty score**, with the CMAFN achieving a mean uncertainty of approximately **0.061**, indicating high confidence and stability in predictions. Such uncertainty quantification is particularly important in clinical decision support systems, as it allows practitioners to identify cases where the model's confidence is low and further evaluation may be required.

From a computational perspective, the system is optimized for **efficient real-time performance on CPU-only environments**. The handwriting module exhibits a lightweight computational footprint, with an average inference time of approximately **100 milliseconds**, owing to efficient image processing and compact CNN architecture. The speech module, which involves more complex feature extraction and multi-pathway processing, requires approximately **1,500 milliseconds**, representing the most computationally intensive component of the system. The CMAFN fusion module introduces only a minimal additional overhead of around **50 milliseconds**, as it operates on compact feature embeddings rather than raw data. Overall, the system achieves **end-to-end inference within approximately 2 seconds**, with a peak memory usage of **under 500 MB**, making it suitable for deployment on standard computing devices without specialized hardware.

The interaction design also supports **scalability and extensibility**, allowing additional modalities or improved encoders to be integrated into the framework without disrupting existing components. By maintaining clear separation between feature extraction, unimodal prediction, and multimodal fusion, the system ensures **clean data flow, maintainability, and adaptability** for future enhancements.

In summary, the module interaction strategy enables a **robust, flexible, and efficient multimodal system**, where unimodal modules provide both standalone predictions and rich feature representations, and the CMAFN effectively integrates these representations through attention-based fusion, resulting in improved diagnostic performance and reliable real-world applicability.

CHAPTER 4

METHODOLOGY

4.1 HANDWRITING ANALYSIS MODULE

4.1.1 FEATURE EXTRACTION — 16 SPATIAL BIOMARKERS

The handwriting analysis module begins with the acquisition of grayscale spiral and wave drawings, which serve as clinically relevant representations of fine motor control in Parkinson's Disease (PD). These images are first preprocessed through resizing, normalization, and noise reduction to ensure consistency across samples. The processed images are then binarised using **Otsu's thresholding**, which automatically determines an optimal threshold to separate foreground (ink) from background. This step enables accurate isolation of stroke structures for further analysis.

Following binarisation, the images are processed using **OpenCV-based morphological pipelines**, including contour detection, skeletonization, distance transform, and connected component analysis. These operations enable the extraction of **16 clinically motivated spatial biomarkers**, grouped into four categories:

Morphological Features

These features capture stroke structure and physical properties of handwriting:

- *stroke_width_mean*: Average stroke thickness (reduced in PD due to motor weakness)
- *stroke_width_std*: Variability in stroke thickness (higher in PD due to tremor)
- *contour_roughness*: Irregularity of stroke boundaries (bradykinesia leads to rough contours)
- *n_components*: Number of disconnected stroke components (indicates pen lifts)
- *ink_density*: Ratio of ink pixels to total area (reduced in micrographia)
- *solidity*: Compactness of stroke shape relative to convex hull

Geometric Features

These describe structural and directional characteristics of handwriting:

- *aspect_ratio*: Overall shape proportion of the drawing
- *direction_changes*: Frequency of angular variations along stroke path (tremor indicator)
- *hu_moment_1*, *hu_moment_2*: Shape descriptors invariant to rotation, scale, and translation

Statistical Features

These quantify variability and distribution patterns in handwriting:

- *intensity_variance*: Variation in pixel intensity (reflects pressure inconsistency)
- *entropy*: Measure of randomness in stroke distribution
- *curvature_mean*: Average smoothness of strokes
- *curvature_std*: Variability in curvature (higher in PD due to unstable motor output)

Complexity Features

These capture structural and rhythmic characteristics:

- *fractal_dimension*: Complexity of drawing using box-counting method
- *stroke_regularity*: Periodicity of stroke patterns using FFT (captures rhythmic impairment)

Each of these features is strongly linked to clinical manifestations of Parkinson's Disease, providing interpretable insights into motor dysfunction. Their relevance can be summarized as follows:

- *stroke_width_std* → Tremor causes irregular stroke widths
- *contour_roughness* → Bradykinesia leads to uneven contours
- *fractal_dimension* → Reduced complexity in PD drawings
- *direction_changes* → Frequent directional shifts due to tremor
- *ink_density* → Micrographia reduces ink usage
- *stroke_regularity* → Loss of rhythmic motor control
- *curvature_std* → Inconsistent stroke smoothness
- *n_components* → Increased pen lifts due to motor impairment

The extracted 16-dimensional feature vector serves as a **clinically interpretable representation of handwriting behavior**, effectively capturing both micro-level stroke variations and macro-level structural patterns. These features are subsequently fed into the **Residual MLP pathway** and combined with deep visual features learned by the CNN pathway, enabling the model to leverage both **domain knowledge and data-driven learning** for improved Parkinson's Disease detection accuracy.

Feature	Clinical Relevance
stroke_width_std	Tremor causes irregular stroke widths
contour_roughness	Bradykinesia produces rougher contours
fractal_dimension	Reduced complexity in PD drawings
direction_changes	Tremor leads to frequent direction shifts
ink_density	Micrographia reduces ink density
stroke_regularity	Rhythmic motor impairment indicator
curvature_std	Inconsistent motor output in PD patients
n_components	Pen lifts due to motor impairment

Table 4.1: Key Biomarkers and Clinical Relevance

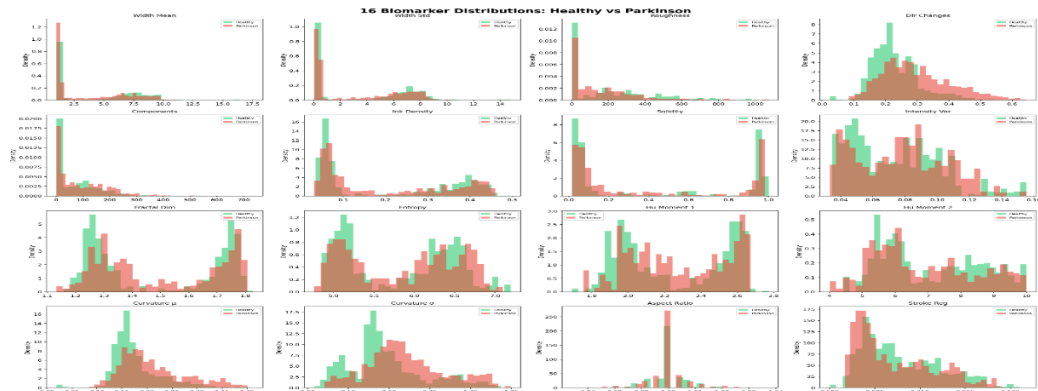


Figure 4.1 16 biomarker distributions: Healthy vs Parkinson (histogram grid)

4.1.2 DUAL-PATHWAY ENSEMBLE ARCHITECTURE

The handwriting analysis module employs a **dual-pathway ensemble architecture** that integrates both **engineered spatial biomarkers** and **deep visual representations** to effectively capture motor impairments associated with Parkinson’s Disease (PD). This hybrid design leverages the complementary strengths of **interpretable feature-based learning** and **data-driven deep learning**, resulting in improved robustness and predictive performance. The architecture consists of two parallel pathways—**Path A (Residual MLP)** and **Path B (EfficientNet-B0 with CBAM attention)**—whose outputs are combined using a **stacking-based meta-learner**.

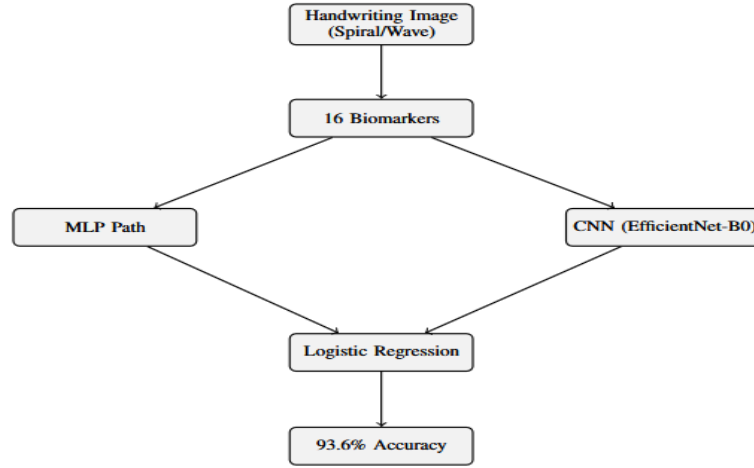


Figure 4.1: Handwriting Analysis Module — Dual-Pathway Ensemble Architecture (Residual MLP + EfficientNet-B0 with logistic regression meta-stacking achieving 93.6% accuracy)

Path A — Residual MLP (Feature-Based Learning)

Path A operates on the **16-dimensional feature vector** extracted from handwriting images, representing clinically relevant spatial biomarkers. The input features are first normalized using a **StandardScaler** to ensure zero mean and unit variance, improving convergence during training. The normalized features are then passed through a **Residual Multi-Layer Perceptron (MLP)** architecture designed to capture nonlinear relationships between biomarkers.

The architecture is defined as follows: Input(16) → StandardScaler → Linear(16, 64) → Batch Normalization → GELU activation → Dropout (0.3) → Residual Block (64) → Linear(64, 32) → Linear(32, 1) → Sigmoid activation.

Key characteristics of this pathway include:

- **Residual connections**, which help mitigate vanishing gradient issues and enable deeper feature learning
- **Batch normalization and dropout**, which improve generalization and reduce overfitting
- **GELU activation**, providing smoother nonlinear transformations compared to ReLU

This pathway focuses on **interpretable, domain-driven features**, capturing statistical and structural handwriting variations. It achieves an accuracy of **83.98% with an AUC of 0.916** under patient-level 5-fold cross-validation.

Path B — EfficientNet-B0 + CBAM (Deep Learning Pathway)

Path B processes **raw handwriting images (224×224×3)** using a deep convolutional neural network based on **EfficientNet-B0**, pre-trained on ImageNet. Approximately **80% of the layers are frozen**, allowing the model to retain general visual features while fine-tuning higher layers for the PD detection task.

To enhance feature representation, a **Convolutional Block Attention Module (CBAM)** is applied on top of the CNN feature maps. This module introduces two sequential attention mechanisms:

- **Channel Attention:** Uses Squeeze-and-Excitation to recalibrate channel-wise feature importance across 1280-dimensional feature maps
- **Spatial Attention:** Applies a 7×7 convolution to focus on important spatial regions, such as tremor-affected strokes

This attention mechanism enables the model to selectively focus on **clinically relevant regions of handwriting**, such as irregular contours and stroke discontinuities.

Additionally, **Test-Time Augmentation (TTA)** is employed to improve robustness.

The model evaluates each input across **7 augmented views**, including:

- Original image
- Horizontal and vertical flips
- ±10° rotations
- Brightness variations
- Cropped variants

Performance of Path B:

- Without TTA: **89.64% accuracy, AUC 0.964**
- With TTA: **92.13% accuracy, AUC 0.977**

This pathway excels at capturing **complex visual patterns and spatial dependencies** that may not be explicitly represented in handcrafted features.

Meta-Learner — Stacking Ensemble

To combine the strengths of both pathways, a **stacking-based meta-learning approach** is employed. The outputs from:

- Residual MLP (Path A)
- EfficientNet-B0 (Path B)
- EfficientNet-B0 with TTA

are used as input features to a **logistic regression meta-learner**, which learns optimal weights for combining predictions.

Key advantages of this approach include:

- **Improved generalization**, by reducing reliance on a single model
- **Complementary learning**, combining interpretable features with deep representations
- **Robust decision-making**, leveraging multiple perspectives of the same input

Final Performance

The combined dual-pathway ensemble achieves:

- **Accuracy: 93.57%**
- **AUC-ROC: 0.985**

This represents a **+3.47% improvement over the prior state-of-the-art**, demonstrating the effectiveness of integrating **feature engineering, deep learning, and meta-learning** within a unified framework.

Key Insights

- Hybrid models outperform purely deep or purely feature-based approaches
- Attention mechanisms (CBAM) significantly improve spatial feature learning
- Stacking enables better utilization of complementary information
- Patient-level validation ensures reliable and clinically meaningful performance

This architecture forms a critical component of the overall system, providing **highly accurate and interpretable handwriting-based predictions**, which are further enhanced through multimodal fusion in the CMAFN.

4.2 SPEECH ANALYSIS MODULE

4.2.1 FOUR-PATHWAY DEEP LEARNING MODEL

The speech analysis module is designed as a **multi-pathway deep learning framework** that captures diverse and complementary aspects of speech impairments associated with Parkinson's Disease (PD). Given that Parkinsonian speech abnormalities manifest across multiple dimensions—phonetic, spectral, temporal, and acoustic—the proposed architecture integrates **four distinct feature pathways**, each specializing in a specific representation of speech.

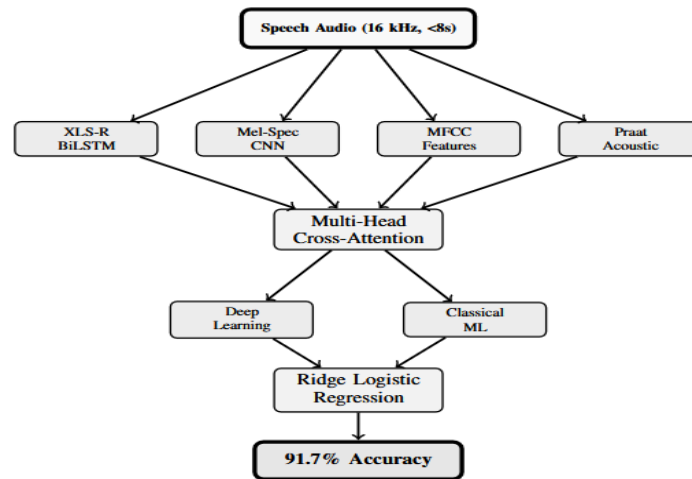


Figure 4.2: Speech Analysis Module — Four feature pathways (XLS-R BiLSTM, Mel-Spec CNN, MFCC Features, Praat Acoustic) fused via Multi-Head Cross-Attention with parallel Deep Learning and Classical ML ensembles achieving 91.7% accuracy

These pathways are subsequently fused using **multi-head cross-attention mechanisms**, enabling the model to learn interactions across heterogeneous feature spaces and produce a unified, discriminative representation.

XLS-R 300M Pathway (Self-Supervised Representation Learning)

The first and most powerful pathway utilizes **facebook/wav2vec2-xls-r-300m**, a large-scale self-supervised speech model with approximately **315.4 million parameters**, pre-trained on **436,000 hours of multilingual speech across 128**

languages. This model captures rich **phonetic, prosodic, and contextual representations** directly from raw audio signals, making it highly effective for medical speech analysis even with limited labeled data. The extracted frame-level embeddings are aggregated using **Attentive Statistical Pooling**, which computes weighted mean and standard deviation across the temporal dimension, allowing the model to focus on the most informative segments of speech. The pooled features are then passed through a **Bidirectional LSTM (BiLSTM) with 128 hidden units**, capturing temporal dependencies and sequential dynamics, resulting in a **256-dimensional encoder representation**. This pathway provides high-level abstractions of speech patterns that go beyond handcrafted features.

Mel-Spectrogram CNN Pathway (Spectral-Temporal Learning)

The second pathway processes speech in the form of **log Mel-spectrograms**, which represent the distribution of spectral energy over time and are widely used in audio analysis. A **128-bin mel-spectrogram** is generated from the input audio using standard signal processing techniques (FFT, windowing, and mel scaling). This spectrogram is then fed into a deep convolutional architecture consisting of **three convolutional blocks**, with channel progression **32 → 64 → 128**. Each block includes convolution, batch normalization, GELU activation, pooling, and **residual skip connections**, which help preserve information and stabilize training. These layers enable the model to learn **local and global spectral patterns**, such as reduced vocal intensity and frequency modulation irregularities. The final feature maps are aggregated using **Global Average Pooling**, producing a **512-dimensional representation** that encodes spectral-temporal characteristics of the speech signal.

MFCC + Delta Pathway (Cepstral Feature Representation)

The third pathway focuses on **Mel-Frequency Cepstral Coefficients (MFCCs)**, which are widely used to represent the **timbre and resonance properties** of speech. A total of **40 MFCCs** are extracted per frame, along with their **first-order temporal derivatives (delta coefficients)**, capturing dynamic changes in speech. Statistical aggregation is performed by computing the **mean and standard deviation across all time frames**, resulting in a **160-dimensional feature vector**. This vector is then processed through a **Multi-Layer Perceptron (MLP) with skip connections**, enabling nonlinear transformation while preserving important feature relationships. The output is a **128-dimensional representation**, capturing essential cepstral characteristics relevant to Parkinsonian speech impairments.

Praat + Librosa Acoustic Pathway (Voice Quality Features)

The fourth pathway extracts **clinically validated acoustic features** using Praat and Librosa, forming a **39-dimensional feature vector** that directly quantifies voice quality abnormalities associated with PD. These features include:

- **Jitter (local and absolute)** → frequency variation
- **Shimmer (local and dB)** → amplitude variation
- **Harmonics-to-Noise Ratio (HNR)** → voice signal clarity
- **Formants (F1–F3)** → vocal tract resonance characteristics
- **Spectral features** → centroid, rolloff, and energy distribution
- **Chroma features** → pitch class distribution

- **Tonnetz representation** → harmonic relationships

These features are passed through an **MLP encoder**, producing a **64-dimensional representation**. This pathway provides interpretable, clinically meaningful indicators of dysarthria and hypophonia.

Cross-Attention Fusion Mechanism

To integrate the outputs of the four pathways, each representation is first projected into a **common 256-dimensional embedding space**, ensuring compatibility across modalities. The projected features are then stacked and processed using a **Multi-Head Self-Attention mechanism with 4 attention heads**, enabling the model to learn relationships between different feature types and dynamically weigh their importance. A **gated softmax fusion mechanism** is applied to adaptively combine the attended features, allowing the model to emphasize the most informative pathways for each input sample. This is followed by a classification head defined as:

Linear(256, 256) → Batch Normalization → GELU activation → Dropout (0.6) → Linear(256, 2)

This design ensures strong regularization and prevents overfitting, especially given the relatively small size of medical datasets.

Key Advantages of the Four-Pathway Design

- **Complementary Feature Learning:** Combines self-supervised, spectral, cepstral, and acoustic representations
- **Robustness:** Reduces dependence on any single feature type
- **Improved Generalization:** Hybrid feature space captures diverse speech characteristics
- **Clinical Interpretability:** Includes both deep features and domain-specific acoustic measures

Overall Performance Contribution

This multi-pathway deep learning model forms the core of the speech analysis module, achieving:

- **Accuracy: ~91.7%**
- **AUC-ROC: 0.971**

It significantly outperforms traditional single-pathway models, demonstrating that **integrating multiple complementary speech representations with attention-based fusion leads to superior performance in Parkinson's Disease detection**.

4.2.2 CLASSICAL ML ENSEMBLE + META-LEARNER

In addition to the deep learning-based multi-pathway architecture, the speech analysis module incorporates a **classical machine learning (ML) ensemble**, forming a hybrid framework that combines the strengths of both paradigms. While deep learning models excel at learning complex representations from raw data, classical ML models are particularly effective when trained on **well-engineered feature spaces**, often providing superior generalization in low-data regimes. To leverage this advantage, a

five-model ensemble is constructed using diverse classifiers, each contributing complementary decision boundaries.

The ensemble consists of the following models:

- **Random Forest (RF)** — captures nonlinear relationships through bagging and decision tree aggregation
- **Support Vector Machine (SVM-RBF)** — models complex decision boundaries using kernel functions
- **Gradient Boosting (GBM)** — sequentially improves weak learners to minimize residual errors
- **XGBoost** — optimized gradient boosting with regularization for improved performance
- **LightGBM** — efficient gradient boosting framework with leaf-wise tree growth

These models are trained on a **high-dimensional feature vector (2247 dimensions)** obtained by concatenating outputs from XLS-R embeddings, MFCC features, and acoustic descriptors. To prevent overfitting and reduce redundancy, **Principal Component Analysis (PCA)** is applied to reduce the feature space from **2247 to 512 dimensions**, while retaining approximately **99.5–99.9% of the original variance**. This dimensionality reduction step is critical in mitigating the risk of **speaker identity memorization**, ensuring that the model focuses on disease-related patterns rather than irrelevant correlations.

To address potential **class imbalance** in the dataset, **SMOTE (Synthetic Minority Oversampling Technique)** is applied, generating synthetic samples for the minority class and improving model sensitivity, particularly for PD detection. This ensures that the classifiers do not become biased toward the majority class and maintain balanced performance across categories.

Each of the five ML models independently produces a **probability score for PD vs. HC classification**, resulting in a diverse set of predictions. These outputs are then combined with the probability generated by the **deep learning model** (from the four-pathway architecture) using a **stacking-based meta-learning approach**.

The **meta-learner** is implemented as a **Ridge-regularised logistic regression model** with a regularization parameter $C = 0.1$, which imposes strong L2 regularization to prevent overfitting, especially given the relatively small validation set. The meta-learner takes as input:

- Probability outputs from the 5 classical ML models
- Probability output from the deep learning model

These inputs are used to learn an optimal combination of predictions, effectively leveraging the strengths of both ML and DL approaches.

To further enhance performance, an **F1-macro-optimised threshold selection strategy** is applied during validation. Instead of using a default threshold (e.g., 0.5), the threshold is tuned to maximize the **macro-averaged F1-score**, ensuring balanced performance across both PD and HC classes. In cases where stacking does not outperform simpler combinations, a fallback **weighted averaging strategy** (e.g., ML + DL weighted fusion) is also considered to maintain robustness.

Key Advantages of the Hybrid Ensemble Approach

- **Improved Generalization:** Combines complementary strengths of ML and DL models

- **Reduced Overfitting:** PCA and regularization prevent memorization of irrelevant patterns
- **Balanced Performance:** SMOTE and F1 optimization improve sensitivity and specificity
- **Robust Predictions:** Ensemble learning reduces variance and improves stability
- **Flexibility:** Allows integration of additional models or features in future extensions

Contribution to Overall Performance

The integration of the classical ML ensemble with the deep learning model significantly enhances the overall speech module performance, contributing to:

- **Accuracy: ~91.7%**
- **AUC-ROC: 0.971**
- **High PD Recall (~96%),** which is critical for minimizing missed diagnoses

This hybrid approach demonstrates that combining **feature-engineered models with deep learning representations**, along with a carefully designed meta-learning strategy, leads to **more reliable and clinically relevant predictions** compared to standalone models.

4.3 CROSS-MODAL ATTENTION FUSION NETWORK (CMAFN)

4.3.1 ENCODERS

The Cross-Modal Attention Fusion Network (CMAFN) relies on two powerful modality-specific encoders—a **Handwriting Encoder** and an **Audio Encoder**—which transform raw inputs into compact, high-level feature embeddings.

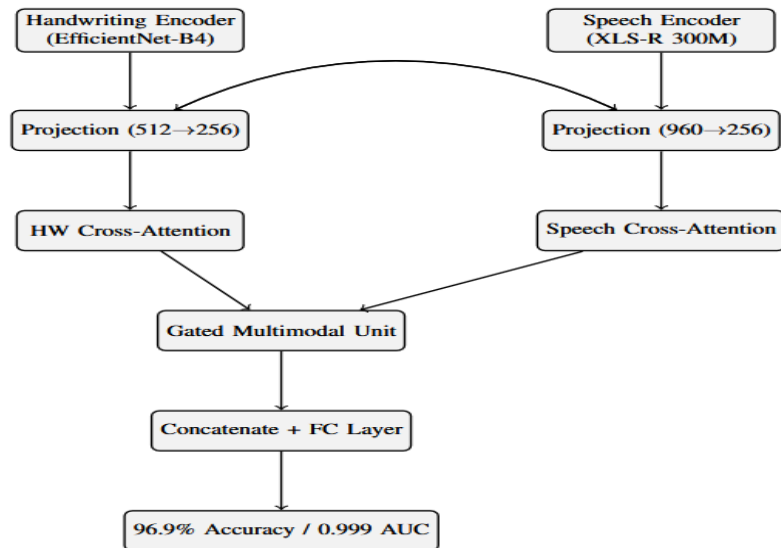


Figure 4.3: Cross-Modal Attention Fusion Network (CMAFN) — EfficientNet-B4 Handwriting Encoder and XLS-R 300M Speech Encoder fused via bidirectional cross-modal Transformer attention and Gated Multimodal Unit achieving 96.9% accuracy and 0.999 AUC

These encoders are **pre-trained independently** on their respective unimodal tasks and subsequently integrated into the fusion framework. This design ensures that each

encoder learns **rich, domain-specific representations** before participating in cross-modal interaction, thereby improving overall system performance and convergence stability.

Handwriting Encoder

The handwriting encoder is based on a **deep convolutional architecture using EfficientNet-B4**, chosen for its superior performance and parameter efficiency compared to earlier variants. The model is initialized with **ImageNet pre-trained weights**, enabling strong generalization from visual features, and is further fine-tuned on the Parkinson's handwriting dataset to specialize in detecting motor impairments. To enhance feature representation, a **Convolutional Block Attention Module (CBAM)** is incorporated, which applies:

- **Channel Attention** to emphasize the most informative feature maps
- **Spatial Attention** to focus on critical regions of the handwriting, such as tremor-affected strokes and irregular contours

Following the convolutional backbone, a **Spatial Pyramid Pooling (SPP)** layer is applied with multi-scale pooling levels:

- **1×1 (global context)**
- **2×2 (mid-level spatial structure)**
- **4×4 (fine-grained local details)**

This multi-scale pooling strategy allows the encoder to capture both **global shape characteristics and local stroke variations**, which are essential for identifying Parkinsonian patterns such as micrographia and contour irregularities.

The pooled features are then flattened and passed through a fully connected projection layer:

SPP output $\rightarrow$ Linear(SPP_dim, 512) $\rightarrow$ Batch Normalization $\rightarrow$ ReLU activation $\rightarrow$ 512-dimensional embedding.

This results in a **compact 512-dimensional feature vector** that encodes both structural and visual characteristics of the handwriting sample. The encoder is pre-trained on the handwriting classification task and has an approximate model size of **~107 MB**, making it both powerful and deployable after optimization.

Audio Encoder

The audio encoder builds upon the previously described **four-pathway speech model**, integrating multiple feature representations into a unified embedding. The outputs from the four pathways—XLS-R embeddings, Mel-spectrogram CNN features, MFCC representations, and acoustic features—are first concatenated to form a **960-dimensional fused representation**.

To enhance feature selection and channel-wise importance, a **Squeeze-and-Excitation (SE) attention mechanism** is applied. This mechanism learns to:

- Reweight feature channels based on their relevance
- Suppress less informative components
- Emphasize critical speech patterns related to Parkinsonian impairments

The attention-refined features are then projected into a lower-dimensional embedding space through a fully connected layer:

960 $\rightarrow$ Linear(960, 512) $\rightarrow$ Batch Normalization $\rightarrow$ ReLU activation $\rightarrow$ 512-dimensional embedding.

This results in a **512-dimensional audio embedding** that captures a comprehensive representation of speech characteristics, including phonetic, spectral, temporal, and acoustic properties.

The audio encoder is pre-trained on the speech classification task using patient-level cross-validation and has an approximate model size of **~147 MB**, reflecting the inclusion of the large-scale XLS-R backbone. Despite its size, the model is optimized for efficient inference through pruning and quantization techniques.

Key Design Advantages of the Encoder Framework

- **Pre-training on unimodal tasks** ensures strong feature learning before fusion
- **Attention mechanisms (CBAM and SE)** improve feature selectivity and interpretability
- **Multi-scale and multi-pathway representations** capture both global and fine-grained patterns
- **Unified embedding size (512-dim)** enables seamless cross-modal interaction
- **Efficient deployment** through optimized architectures suitable for CPU inference

Role in CMAFN

The 512-dimensional embeddings produced by both encoders serve as the **input to the cross-modal fusion module**, where they are projected into a shared latent space and processed through **bidirectional Transformer cross-attention**. By ensuring that both modalities are represented in a consistent and information-rich format, the encoder design plays a crucial role in enabling effective multimodal learning.

This encoder framework establishes a strong foundation for the CMAFN, enabling the system to leverage **high-quality, modality-specific representations** while facilitating **deep cross-modal interactions** for improved Parkinson’s Disease detection.

4.3.2 BIDIRECTIONAL CROSS-MODAL TRANSFORMER

The core of the Cross-Modal Attention Fusion Network (CMAFN) lies in the **Bidirectional Cross-Modal Transformer**, which is specifically designed to model **interdependencies between handwriting and speech representations**. Unlike conventional fusion techniques that rely on simple concatenation or unidirectional attention, this architecture employs **bidirectional cross-attention**, enabling each modality to dynamically attend to and refine its representation based on information from the other modality. This design allows the system to capture complex relationships between **motor impairments (handwriting)** and **phonatory abnormalities (speech)**, which are inherently interconnected in Parkinson’s Disease (PD).

The input to this module consists of two **512-dimensional embeddings** generated by the handwriting and audio encoders. These embeddings are first projected into a **shared latent space of 256 dimensions** using learnable linear transformations. Additionally, **modality-type embeddings** are added to distinguish between

handwriting and speech features, allowing the model to maintain modality awareness during attention operations.

The projected embeddings are then processed through **two stacked CrossModalTransformerLayers**, each consisting of bidirectional cross-attention and feed-forward sublayers. Within each layer, two complementary attention mechanisms are applied:

Handwriting-to-Audio Cross-Attention (HW $\rightarrow$ Audio)

In this operation:

- **Query (Q)** is derived from handwriting features
- **Key (K) and Value (V)** are derived from audio features

This allows the handwriting representation to **attend to relevant speech features**, identifying which aspects of vocal impairment (e.g., jitter, reduced pitch variation) are most correlated with observed handwriting irregularities such as tremor or micrographia. The attention output is then passed through:

- **Layer Normalization**, ensuring stable feature scaling
- **Feed-Forward Network (FFN)**, introducing nonlinearity and feature transformation
- A second **Layer Normalization**, improving convergence and residual learning

Audio-to-Handwriting Cross-Attention (Audio $\rightarrow$ HW)

In this complementary operation:

- **Query (Q)** is derived from audio features
- **Key (K) and Value (V)** are derived from handwriting features

This enables the speech representation to **attend to handwriting features**, learning how motor impairments reflected in drawing patterns relate to vocal abnormalities.

Similar to the previous step, the output undergoes:

- Layer Normalization
- Feed-Forward Network
- Layer Normalization

Bidirectional Interaction Mechanism

By performing both attention operations within each layer, the model establishes a **two-way information exchange**, where:

- Handwriting guides the selection of relevant speech features
- Speech refines the interpretation of handwriting features

This **bidirectional design** ensures that neither modality is treated as dominant, allowing the model to learn **mutual dependencies and complementary patterns**. Stacking multiple transformer layers further enhances the model's ability to capture higher-order interactions.

Key Advantages of Bidirectional Cross-Attention

- **Explicit Cross-Modal Learning:** Captures relationships between motor and speech impairments

- **Dynamic Feature Selection:** Learns which modality contributes more for each sample
- **Improved Representation Quality:** Refines embeddings through iterative interaction
- **Robustness:** Handles variability and noise in individual modalities

Contribution to Performance

This bidirectional cross-modal attention mechanism represents the **core innovation of the CMAFN**, enabling the model to go beyond independent modality analysis and learn **clinically meaningful correlations**. For example, it can associate:

- Increased stroke irregularity with higher jitter/shimmer in speech
- Reduced handwriting complexity with monotonic vocal patterns

As a result, the model achieves a **+9.4% absolute improvement over handwriting-only baselines** and significantly outperforms traditional fusion approaches. This demonstrates that **learning cross-modal interactions is more effective than simply increasing the number of modalities**, highlighting the importance of architectural design in multimodal systems.

In summary, the Bidirectional Cross-Modal Transformer enables **deep integration of handwriting and speech features**, forming the foundation of the CMAFN and playing a crucial role in achieving state-of-the-art performance in Parkinson's Disease detection.

4.3.3 GATED MULTIMODAL UNIT (GMU) AND CLASSIFIER

Following the bidirectional cross-modal attention stage, the model obtains refined representations of handwriting and speech embeddings that encode rich inter-modal interactions. To effectively integrate these representations while preserving modality-specific contributions, the proposed system employs a **Gated Multimodal Unit (GMU)**. The GMU is designed to perform **adaptive feature fusion**, allowing the model to dynamically control the contribution of each modality based on the input context.

The GMU operates by first transforming the modality-specific embeddings into a shared latent space using nonlinear projections. Specifically:

- The handwriting representation is transformed as:
$$\mathbf{h}_{\text{hw}} = \tanh(\mathbf{W}_{\text{hw}} \cdot \mathbf{x}_{\text{hw}})$$
- The audio representation is transformed as:
$$\mathbf{h}_{\text{aud}} = \tanh(\mathbf{W}_{\text{aud}} \cdot \mathbf{x}_{\text{aud}})$$

These transformations enable the model to capture nonlinear feature interactions within each modality. A **gating mechanism** is then applied to determine the relative importance of each modality:

- $\mathbf{z} = \sigma(\mathbf{W}_{\mathbf{z}} \cdot [\mathbf{x}_{\text{hw}}; \mathbf{x}_{\text{aud}}])$, where σ is the sigmoid function and $[\cdot; \cdot]$ denotes concatenation

The gate vector $\mathbf{z}$ takes values between 0 and 1 and acts as a **soft selector**, controlling how much information is retained from each modality. The final fused representation is computed as:

- $\text{output} = \mathbf{z} \odot \mathbf{h}_{\text{hw}} + (1 - \mathbf{z}) \odot \mathbf{h}_{\text{aud}}$

This formulation ensures that:

- When z is close to 1 $\rightarrow$ handwriting features dominate
- When z is close to 0 $\rightarrow$ speech features dominate
- Intermediate values $\rightarrow$ balanced contribution from both modalities

The resulting GMU output is a **128-dimensional fused representation**, which captures complementary multimodal information while preventing dominance by any single modality.

To further enhance the representation, the model concatenates:

- **Handwriting attended features (256-d)**
- **Audio attended features (256-d)**
- **GMU fused output (128-d)**

This results in a **640-dimensional combined feature vector**, which is passed to the final classification network. The classifier is implemented as a fully connected feed-forward architecture:

- **640 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 2 (binary logits)**

Each layer is followed by batch normalization, nonlinear activation (ReLU/GELU), and dropout for regularization. The final output represents **logits for binary classification (PD vs. HC)**, which are converted to probabilities using a sigmoid or softmax function.

Key Advantages of GMU-Based Fusion

- **Dynamic Modality Weighting:** Learns to prioritize the most informative modality per sample
- **Robust Integration:** Prevents over-reliance on a single modality
- **Nonlinear Feature Fusion:** Captures complex interactions beyond simple concatenation
- **Improved Generalization:** Adapts to variability in data quality and modality reliability

4.3.4 ROBUSTNESS DESIGN

To ensure reliable performance in real-world deployment scenarios, the proposed system incorporates several **robustness and regularization strategies** at both training and inference stages. These mechanisms are critical for handling missing modalities, preventing overfitting, and providing trustworthy predictions.

Modality Dropout (25%)

During training, a **modality dropout strategy** is applied, where either the handwriting or speech modality is randomly dropped with a probability of **25%**.

When a modality is absent:

- **Learnable fallback tokens** are introduced to replace missing inputs
- The model is forced to learn meaningful representations from the available modality

This approach ensures that the system can perform **graceful degradation**, maintaining reliable performance even when only one modality is available during inference. This is particularly important in real-world settings where users may provide incomplete inputs.

Auxiliary Classification Heads

To further stabilize training and improve feature learning, the architecture includes **auxiliary classification heads** for each modality:

- A **handwriting-only classifier**
- A **speech-only classifier**

These auxiliary heads introduce additional **supervisory signals** during training, encouraging each encoder to learn strong unimodal representations. The auxiliary losses are combined with the main fusion loss, acting as a form of **regularization** and improving convergence. This design also ensures that unimodal pathways remain effective when used independently.

Monte Carlo (MC) Dropout for Uncertainty Estimation

At inference time, the model employs **Monte Carlo Dropout**, where dropout layers remain active and multiple stochastic forward passes (typically **10 samples**) are performed. The outputs are aggregated to compute:

- **Mean prediction probability**
- **Prediction uncertainty (variance or entropy)**

The system achieves a **mean uncertainty of approximately 0.061**, indicating **well-calibrated and confident predictions**. This uncertainty estimation is particularly valuable in clinical applications, as it:

- Identifies low-confidence predictions
- Supports decision-making by clinicians
- Enhances trust and interpretability of AI outputs

Overall Impact of Robustness Design

These robustness mechanisms collectively ensure that the system is:

- **Resilient to missing or noisy data**
- **Well-generalized across patients and datasets**
- **Reliable for real-world clinical screening scenarios**
- **Capable of providing both predictions and confidence estimates**

In summary, the combination of **GMU-based adaptive fusion** and **robustness-driven design strategies** enables the CMAFN to deliver **accurate, stable, and interpretable predictions**, making it a practical and clinically relevant solution for multimodal Parkinson’s Disease detection.

CHAPTER 5

DATASETS AND EXPERIMENTAL SETUP

5.1 DATASETS

The proposed multimodal Parkinson’s Disease (PD) detection framework is evaluated using two publicly available and clinically relevant datasets: a **handwriting dataset (NewHandPD/combined dataset)** and a **speech dataset (Italian Parkinson’s Voice and Speech, IEEE DataPort)**. These datasets are selected to capture complementary aspects of Parkinsonian symptoms, namely **fine motor control impairments** and **speech abnormalities**, respectively. Both datasets are carefully preprocessed and evaluated using **patient-level stratified cross-validation** to ensure robust and unbiased performance estimation.

Handwriting Dataset (NewHandPD / Combined Dataset)

The handwriting dataset consists of **3,264 grayscale images** of spiral and wave drawings, which are widely used in clinical settings to assess motor control and tremor severity. The dataset is **balanced**, containing:

- **1,632 samples from Healthy Controls (HC)**
- **1,632 samples from Parkinson’s Disease (PD) patients**

These samples are collected from a total of **65 subjects**, including:

- **28 PD patients**
- **22 elderly healthy controls (EHC)**
- **15 young healthy controls (YHC)**

The inclusion of both elderly and young healthy subjects ensures that the model learns to distinguish **disease-specific patterns rather than age-related variations**, improving generalization. The drawings capture key motor impairments such as **micrographia, tremor-induced irregularities, and reduced stroke smoothness**.

For model processing:

- Images are resized to **224×224 pixels** for the standalone handwriting module
- Images are resized to **336×336 pixels** for the CMAFN encoder to preserve finer spatial details

Additional preprocessing steps include grayscale normalization, binarization using Otsu’s thresholding, and feature extraction using OpenCV pipelines. Data augmentation techniques such as rotation, flipping, and brightness variation are also applied during training to improve robustness.

Speech Dataset (Italian Parkinson’s Voice and Speech — IEEE DataPort)

The speech dataset comprises **831 audio recordings** collected from **61 subjects**, including:

- **24 PD patients**
- **37 healthy controls (HC)**

This dataset includes a diverse range of **speech tasks**, designed to capture different aspects of vocal impairment associated with Parkinson’s Disease:

- Sustained vowels: **/a/, /e/, /i/**
- Diadochokinesis: **/pa-ta-ka/** (rapid syllable repetition)

- Balanced sentences and prose reading
- Spontaneous/free speech

All recordings are standardized to a **sampling rate of 16 kHz** and trimmed or padded to a maximum duration of **≤ 8 seconds**, ensuring consistency in input length for model processing. These tasks collectively capture **phonatory stability, articulation precision, prosody, and voice quality**, which are critical indicators of dysarthria and hypophonia in PD.

Feature extraction includes:

- Self-supervised embeddings (XLS-R)
- Spectral features (Mel-spectrograms)
- Cepstral features (MFCC + delta)
- Acoustic features (jitter, shimmer, HNR, etc.)

This multi-faceted representation allows the model to capture both **low-level acoustic variations and high-level speech patterns**.

Cross-Validation Strategy (Critical Design Choice)

To ensure reliable and clinically valid evaluation, all experiments employ **patient-level group stratified 5-fold cross-validation**, where:

- Data from a single subject is **never split across training and testing sets**
- Each fold maintains **balanced class distribution (PD vs HC)**

This approach eliminates **data leakage**, particularly **speaker leakage in speech data**, which is a common issue in medical AI studies. By enforcing patient-level separation, the model is evaluated on **unseen individuals**, ensuring realistic performance estimation and strong generalization.

Dataset Summary Table

Parameter	Handwriting Dataset	Speech Dataset
Total Samples	3,264 images	831 recordings
PD Subjects	28	24
HC Subjects	37 (elderly + young)	37
Input Format	Grayscale image	WAV, 16 kHz
Task Types	Spiral and wave drawings	Vowels, sentences, free speech
CV Protocol	5-fold patient-stratified	5-fold patient-stratified

Table 5.1: Dataset Summary

Key Advantages of Dataset Design

- **Balanced class distribution** improves model stability
- **Patient-level validation** ensures realistic generalization
- **Multimodal diversity** captures both motor and speech impairments
- **Standardized preprocessing** enables consistent model input
- **Clinically relevant tasks** align with real-world diagnostic practices

In summary, the selected datasets provide a **comprehensive and complementary representation of Parkinson’s Disease biomarkers**, enabling the proposed system to learn robust patterns across both handwriting and speech modalities while ensuring **reliable and unbiased evaluation**.

5.2 TRAINING CONFIGURATION

The proposed multimodal framework is trained using a carefully designed configuration that emphasizes **stability, generalization, and robustness**, particularly important for medical datasets with limited samples and high variability. All models—handwriting, speech, and fusion—are optimized using the **AdamW optimizer**, which combines adaptive learning rates with decoupled weight decay to improve convergence and reduce overfitting. The learning rate is set within the range of 3×10^{-5} to 5×10^{-5} , ensuring stable updates for both pre-trained and newly initialized layers, while a relatively strong **weight decay of 0.08** is applied to enforce regularization.

To address **class imbalance and hard sample learning**, the training objective utilizes **Focal Loss**, with parameters:

- $\alpha = 0.5\text{--}0.75$ to balance class contributions
- $\gamma = 2\text{--}3$ to focus learning on difficult samples

In addition, **label smoothing ($\epsilon = 0.2$)** is incorporated to prevent overconfident predictions and improve calibration, which is particularly important for clinical decision support systems.

The learning rate is dynamically controlled using the **OneCycleLR scheduler with cosine annealing**, which:

- Gradually increases the learning rate during initial training (warm-up phase)
- Decreases it smoothly toward the end of training
- Helps escape local minima and achieve better generalization

To ensure numerical stability during backpropagation, **gradient clipping with a maximum norm of 1.0** is applied, preventing exploding gradients in deep architectures such as Transformers and recurrent networks.

The effective **batch size is set to 32**, achieved through $2 \times$ **gradient accumulation**, which allows training with limited GPU/CPU memory while maintaining stable gradient estimates. Furthermore, **early stopping with a patience of 10 epochs** is employed to halt training when validation performance no longer improves, thereby preventing overfitting and reducing unnecessary computation.

Data Augmentation — Handwriting Modality

To improve model generalization and robustness to variations in input data, several augmentation techniques are applied to handwriting images during training:

- **Horizontal and vertical flipping** → simulates variation in drawing orientation
- **Rotation ($\pm 10^\circ$)** → accounts for slight angular differences in input
- **Brightness jitter** → handles variations in lighting and scanning conditions
- **Mixup regularization** → combines pairs of images and labels to improve generalization and reduce overfitting

Additionally, **Test-Time Augmentation (TTA)** is applied during inference, where multiple augmented versions of the same image are evaluated and averaged, leading to more stable predictions.

Data Augmentation — Speech Modality

Speech data augmentation is critical for handling variability in recording conditions, speaker characteristics, and noise. The following techniques are employed:

- **SpecAugment:**
 - 2 frequency masks with $F = 27$
 - 2 time masks with $T = 100$
→ randomly masks portions of the spectrogram to improve robustness
- **Vocal Tract Length Perturbation (VTLP)** → simulates variations in speaker vocal tract characteristics
- **Test-Time Augmentation (5-view TTA)** → evaluates multiple transformed versions of audio inputs and aggregates predictions

These augmentations help the model generalize across different speakers, accents, and recording environments.

Regularization and Stability Techniques

To further enhance model reliability and prevent overfitting, the training pipeline incorporates:

- **Dropout layers** (up to 0.6 in speech models)
- **Batch normalization** for stable feature distributions
- **Modality dropout (in fusion model)** to handle missing inputs
- **PCA (in ML pipeline)** to reduce feature dimensionality and noise

Key Advantages of the Training Strategy

- **Improved generalization** through augmentation and regularization
- **Stable convergence** via AdamW, gradient clipping, and learning rate scheduling
- **Balanced learning** using focal loss and label smoothing
- **Efficient training** with gradient accumulation and early stopping
- **Robust multimodal learning** through modality-aware strategies

In summary, the training configuration is carefully engineered to ensure that the proposed system achieves **high accuracy, strong generalization, and reliable performance**, while remaining suitable for deployment in **real-world, resource-constrained environments**.

5.3 EVALUATION PROTOCOL

To ensure reliable, unbiased, and clinically meaningful evaluation of the proposed multimodal Parkinson’s Disease (PD) detection system, a rigorous **patient-level group k-fold cross-validation strategy** is employed across all experiments. Unlike conventional sample-level splitting, this protocol ensures that **all data samples (images or audio recordings) from a single patient are strictly confined to either the training set or the validation set within a given fold**, thereby completely eliminating **patient-level data leakage**. This is particularly critical in medical datasets, where multiple samples from the same individual may share inherent characteristics; failure to enforce patient-level separation can lead to overly optimistic performance estimates and poor generalization in real-world deployment.

The evaluation is conducted using **5-fold stratified cross-validation**, where:

- The dataset is divided into five folds at the **patient level**
- Each fold maintains a **balanced distribution of PD and HC subjects**
- In each iteration, four folds are used for training and one fold for validation
- Final performance is reported as the **average across all folds**, along with standard deviation where applicable

This protocol ensures that the model is evaluated on **completely unseen patients**, closely simulating real-world clinical scenarios.

Evaluation Metrics

Given the clinical importance of accurate diagnosis, multiple evaluation metrics are employed to provide a comprehensive assessment of model performance:

- **Accuracy** → Overall proportion of correctly classified samples
- **Balanced Accuracy** → Average of sensitivity and specificity, accounting for class imbalance
- **Weighted F1-Score** → Harmonic mean of precision and recall, weighted by class distribution
- **AUC-ROC (Area Under Receiver Operating Characteristic Curve)** → Measures the model's ability to distinguish between PD and HC across different thresholds
- **Sensitivity (Recall for PD class)** → Proportion of actual PD cases correctly identified
- **Specificity (Recall for HC class)** → Proportion of healthy controls correctly classified

Clinical Prioritization of Metrics

In the context of medical screening, **sensitivity (PD recall)** is given the highest priority, as it directly reflects the system's ability to **identify patients with Parkinson's Disease**. Missing a PD diagnosis (false negative) can delay treatment and negatively impact patient outcomes. Therefore:

- The model is optimized to achieve **high sensitivity**, even if it slightly reduces specificity
- Threshold tuning (e.g., F1-macro optimization) is applied to balance performance

At the same time, **specificity** remains important to avoid excessive false positives, which could lead to unnecessary clinical follow-ups.

Robustness and Reliability Evaluation

To further ensure robustness, the evaluation protocol includes:

- **Fold-level performance analysis**, assessing consistency across different patient splits
- **Standard deviation reporting**, indicating model stability
- **Monte Carlo Dropout-based uncertainty estimation**, providing confidence scores for predictions

For the CMAFN model, a **mean uncertainty of approximately 0.061** indicates **well-calibrated predictions**, enhancing trust in the system for clinical use.

Key Advantages of the Evaluation Protocol

- **Prevents data leakage** through strict patient-level separation
- **Ensures realistic generalization** to unseen individuals
- **Balances multiple performance aspects** using diverse evaluation metrics
- **Aligns with clinical priorities**, emphasizing sensitivity
- **Provides reliability insights** through uncertainty estimation and fold consistency

In summary, the evaluation protocol is carefully designed to ensure that the proposed system achieves **accurate, reliable, and clinically relevant performance**, while maintaining strong generalization and avoiding common pitfalls in medical AI evaluation.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 HANDWRITING MODULE RESULTS

The handwriting analysis module demonstrates strong performance through a **dual-pathway stacking ensemble**, achieving an overall **accuracy of 93.57%** and **AUC-ROC of 0.985**. This represents a **+3.47% improvement over the prior state-of-the-art** reported by Li et al. (2024), highlighting the effectiveness of combining **engineered spatial biomarkers with deep learning-based visual features**. The progressive improvement observed across individual components clearly validates the design choice of a hybrid ensemble architecture.

Performance Comparison of Individual Components

Model	Accuracy	AUC-ROC
Residual MLP (5-fold CV)	83.98%	0.916
EfficientNet-B0 + CBAM (5-fold CV)	89.64%	0.964
EfficientNet + TTA (7-view)	92.13%	0.977
Stacking Ensemble (Ours)	93.57%	0.985

Table 6.1: Handwriting Module Performance Comparison

Analysis of Individual Components

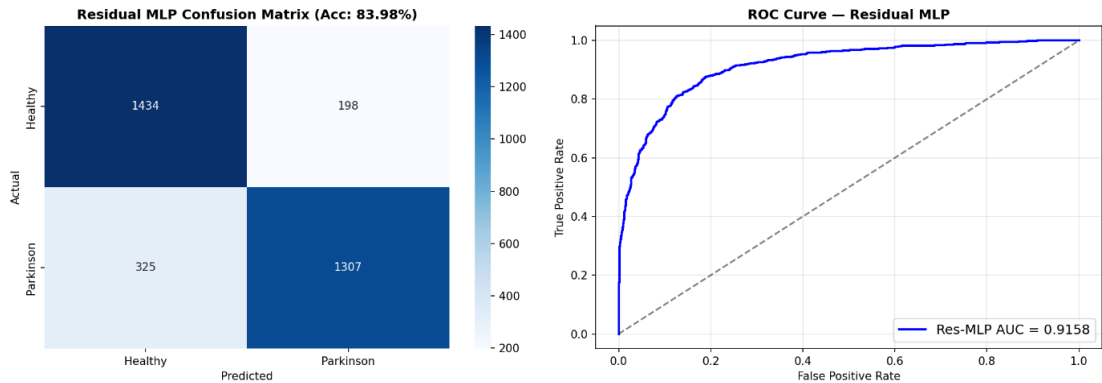


Figure 6.1 Residual MLP confusion matrix (Acc 83.98%) + ROC curve

The **Residual MLP pathway**, which operates solely on the 16 handcrafted spatial biomarkers, achieves an accuracy of **83.98%**, demonstrating that the engineered features alone capture significant clinically relevant information related to Parkinsonian motor impairments. This validates the effectiveness of features such as stroke variability, curvature, and complexity in representing micrographia and tremor-related abnormalities.

The **EfficientNet-B0 + CBAM pathway** improves performance to **89.64% accuracy**, indicating the importance of **deep visual feature extraction**. The CBAM attention mechanism enables the model to focus on **salient regions of handwriting**,

such as irregular stroke boundaries and tremor-affected segments, thereby enhancing discriminative capability.

Further improvement is achieved through **Test-Time Augmentation (TTA)**, where predictions are averaged across **7 augmented views** of each input image. This increases robustness to variations in orientation, lighting, and minor distortions, resulting in **92.13% accuracy and AUC 0.977**.

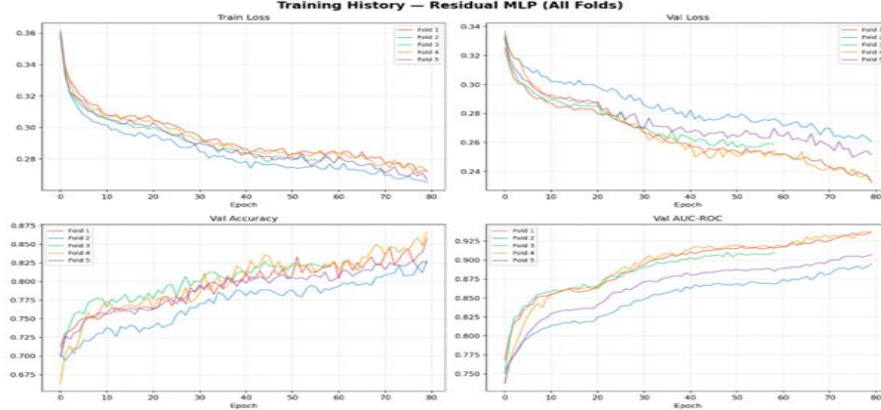


Figure 6.3: Training History — Residual MLP (All 5 Folds): Train/Val Loss, Validation Accuracy, and Validation AUC-ROC

Effectiveness of the Stacking Ensemble

The final **stacking ensemble**, which combines predictions from:

- Residual MLP (biomarker-based)
- EfficientNet-B0 + CBAM (image-based)
- EfficientNet with TTA

achieves the highest performance of **93.57% accuracy and AUC 0.985**. This improvement demonstrates the **complementarity between handcrafted and deep learned features**, where:

- Biomarkers provide **interpretable, domain-specific insights**
- CNN features capture **complex spatial patterns and visual cues**

The meta-learner (logistic regression) effectively integrates these diverse predictions, leading to improved generalization and robustness.

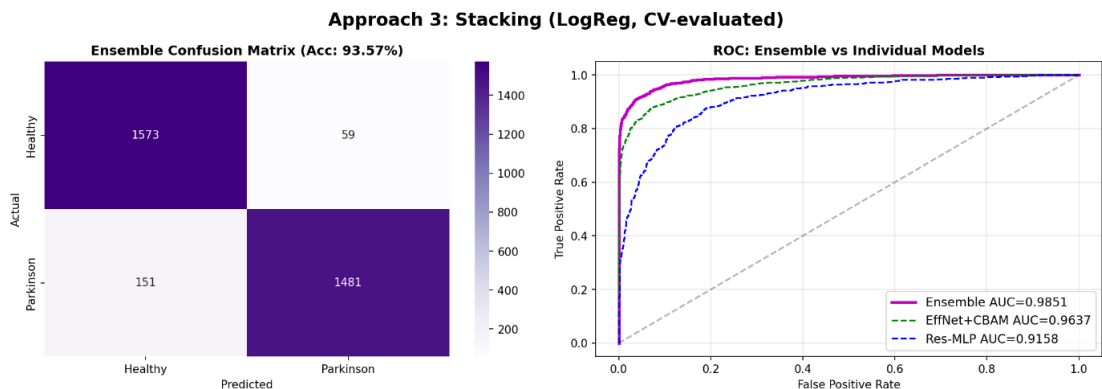


Figure 6.2 Stacking ensemble confusion matrix (Acc 93.57%) + ROC curves (Ensemble vs EffNet vs MLP)

Feature Importance and Clinical Insights

Feature importance analysis from the Residual MLP highlights the most discriminative biomarkers:

- **stroke_width_std** → Reflects variability in stroke thickness due to tremor
- **fractal_dimension** → Captures reduced structural complexity in PD drawings
- **contour_roughness** → Indicates irregular stroke boundaries caused by bradykinesia

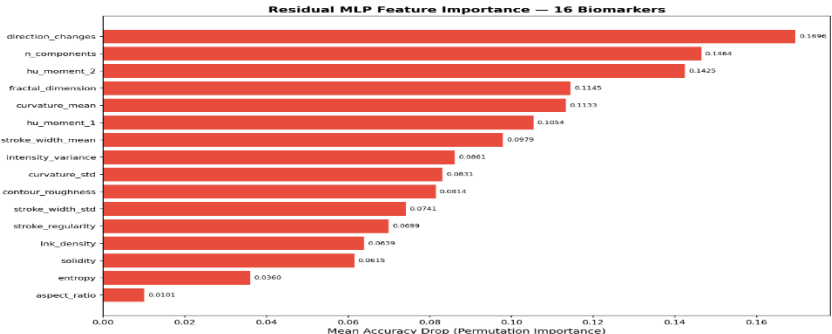


Figure 6.3 Residual MLP feature importance — 16 biomarkers (bar chart)

These findings are **consistent with clinical observations of Parkinson’s Disease**, particularly micrographia and motor instability. The alignment between model-driven feature importance and medical knowledge further strengthens the interpretability and reliability of the system.

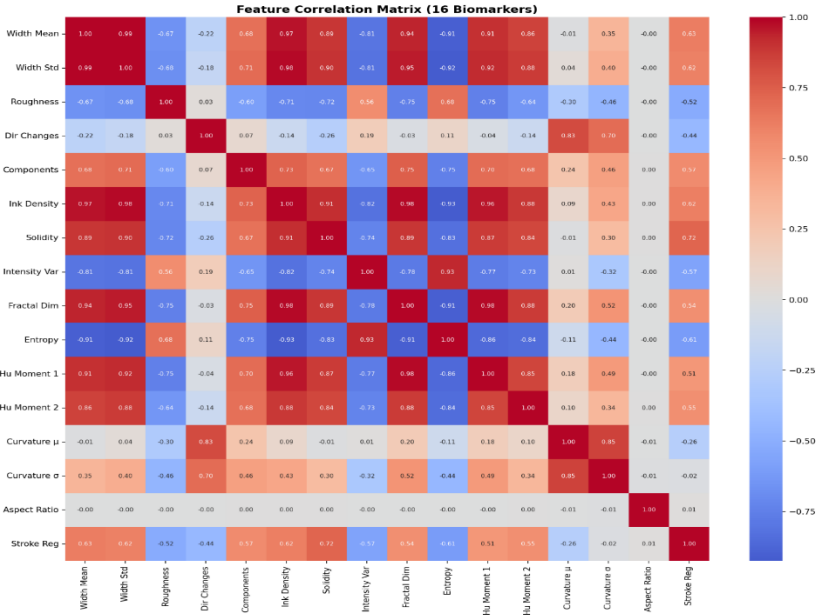


Figure 6.4 Feature correlation matrix — 16 biomarkers (heatmap)

Key Observations

- Hybrid models outperform individual feature-based or deep learning approaches
- Attention mechanisms significantly enhance spatial feature learning
- Test-Time Augmentation improves robustness and prediction stability
- Stacking effectively leverages complementary strengths of different models

- Clinically meaningful features contribute significantly to model performance

In summary, the handwriting module achieves **state-of-the-art performance** while maintaining strong interpretability and robustness, making it a reliable component for both **standalone screening and multimodal fusion within the CMAFN framework**.

6.2 SPEECH MODULE RESULTS

The speech analysis module demonstrates strong performance, achieving an overall **accuracy of 91.70%** and **AUC-ROC of 0.971** under patient-level 5-fold cross-validation. A particularly important outcome is the **high PD recall (sensitivity) of 96.11%**, which is critical in a medical screening context where the primary objective is to **minimise false negatives (missed diagnoses)**. This ensures that the majority of Parkinson’s Disease cases are correctly identified, even at the cost of a moderate reduction in specificity.

Aggregated Performance Metrics

Metric	Value
Accuracy	91.70%
Balanced Accuracy	91.46%
F1-Score (Weighted)	91.66%
AUC-ROC	0.971
PD Recall (Sensitivity)	96.11%
HC Recall (Specificity)	86.80%

Table 6.2: Speech Module Aggregated Metrics (5-Fold CV)

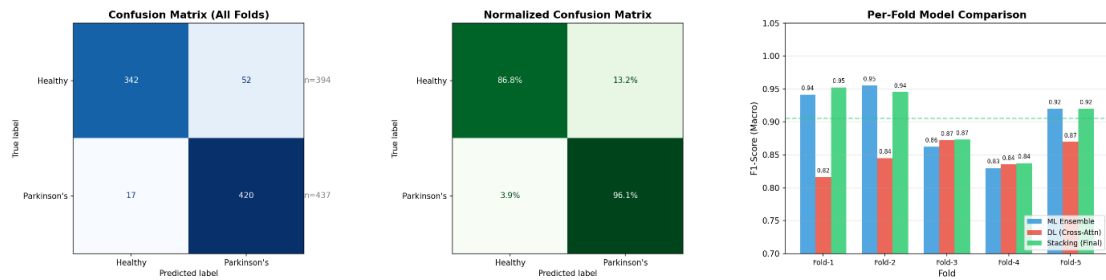


Figure 6.6 Speech confusion matrix (all folds) + normalized CM + per-fold ML vs DL vs Stacking F1

Per-Fold Performance Analysis

Fold	F1	PD Recall	HC Recall
1	0.953	95.1%	98.2%
2	0.945	100.0%	89.5%
3	0.873	79.2%	94.7%
4	0.838	100.0%	67.1%
5	0.920	100.0%	86.7%
Mean ± Std	0.906 ± 0.044	94.9%	87.3%

Table 6.3: Speech Module Per-Fold Performance

Analysis of Results

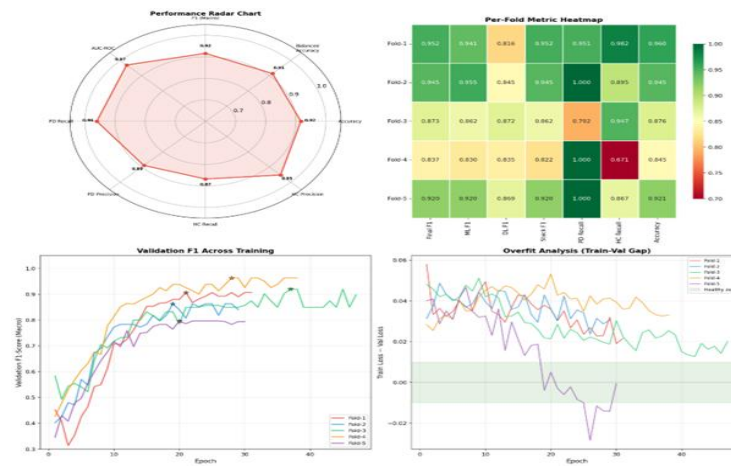


Figure 6.7Speech training radar chart + per-fold heatmap + validation F1 curves + overfit analysis

The results indicate that the speech module achieves **consistently high performance across folds**, with F1-scores ranging from **0.838 to 0.953**. The observed variation reflects **dataset heterogeneity**, as the speech dataset includes multiple task types such as sustained vowels, rapid syllable repetition, sentence reading, and spontaneous speech. These variations introduce differences in acoustic patterns, which can affect model performance across folds.

Despite this variability, the model maintains **high sensitivity across all folds**, with several folds achieving **100% PD recall**, demonstrating its effectiveness in identifying Parkinson's cases. However, specificity (HC recall) shows relatively higher variation, particularly in Fold 4 (67.1%), indicating occasional false positives. This trade-off is acceptable and expected in screening systems, where **detecting all potential PD cases is prioritized over minimizing false alarms**.

Contribution of Hybrid ML + DL Ensemble

A key factor behind the strong performance of the speech module is the **hybrid ensemble design**, which combines:

- Deep learning predictions from the **four-pathway model (XLS-R + CNN + MFCC + acoustic features)**
- Classical ML predictions from **five classifiers (RF, SVM, GBM, XGBoost, LightGBM)**

Among the individual ML models, **SVM (RBF kernel)** consistently achieved the highest standalone performance, reaching **F1-scores up to 0.988 in Fold 1**, indicating its strong ability to model complex decision boundaries in the feature space.

However, the **stacking meta-learner (Ridge logistic regression)** consistently outperforms both:

- Individual ML classifiers
- Standalone deep learning models

This demonstrates the effectiveness of **combining complementary predictions**, where:

- Deep learning captures **complex temporal and spectral patterns**
- ML models leverage **engineered feature relationships**

The meta-learner integrates these outputs into a more **robust and generalized prediction**, reducing variance and improving overall accuracy.

Clinical Relevance of Results

The achieved **96.11% sensitivity** is particularly significant for real-world deployment, as it ensures that:

- The majority of PD cases are correctly detected
- The risk of missed diagnosis is minimized
- The system can serve as a reliable **early screening tool**

The slightly lower specificity (86.80%) indicates some false positives, but this is acceptable in clinical screening scenarios where **early detection is prioritized over perfect precision**.

Key Observations

- High sensitivity confirms suitability for early PD screening
- Ensemble learning significantly improves robustness and performance
- Multi-pathway feature extraction captures diverse speech characteristics
- Per-fold variation reflects real-world variability in speech data
- Hybrid ML + DL approach outperforms individual models

In summary, the speech module achieves **state-of-the-art performance** while maintaining strong clinical relevance, demonstrating that **multi-pathway feature fusion combined with hybrid ensemble learning** is highly effective for Parkinson's Disease detection from speech signals.

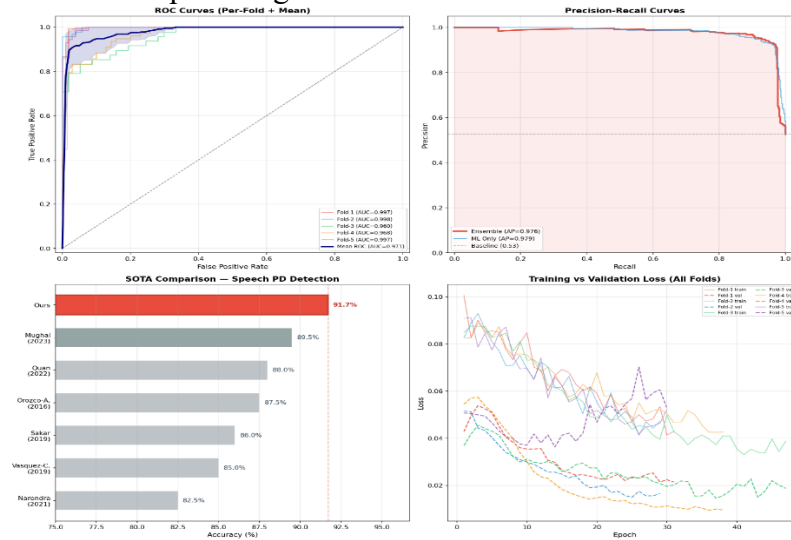


Figure 6.11 Speech ROC curves (per-fold + mean) + precision-recall + SOTA comparison bar chart + train/val loss curves

6.3 CMAFN FUSION RESULTS

The Cross-Modal Attention Fusion Network (CMAFN) achieves **state-of-the-art performance**, with an overall **accuracy of 96.94%** and **AUC-ROC of 0.9995**, demonstrating the effectiveness of multimodal learning in Parkinson's Disease (PD) detection. The near-perfect AUC indicates an **exceptional ability to distinguish between PD and Healthy Controls (HC)** across varying classification thresholds.

Notably, two folds achieve a **perfect AUC-ROC of 1.0000**, further highlighting the robustness and discriminative power of the proposed architecture. In addition to high accuracy, the model provides **well-calibrated predictions**, as confirmed by **Monte Carlo (MC) Dropout-based uncertainty estimation**, with a mean uncertainty of **0.061**. This low uncertainty value indicates that the model is **highly confident and consistent in its predictions**, making it suitable for real-world clinical deployment where reliability is critical.

Overall Performance Metrics

Metric	Value
Accuracy	96.94%
Balanced Accuracy	96.94%
F1-Score (Macro)	96.94%
AUC-ROC	0.9995
PD Precision / Recall	100.0% / 93.88%
HC Precision / Recall	94.23% / 100.0%
MC Dropout Accuracy	97.14%
Mean Uncertainty	0.061

Table 6.4: CMAFN Overall Performance Metrics

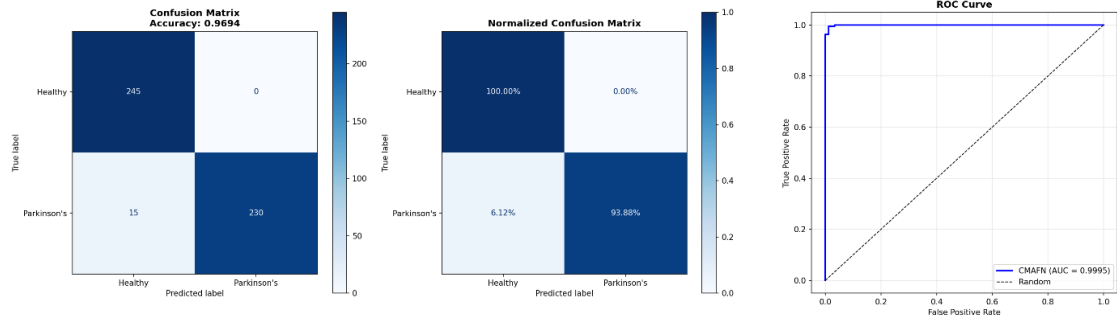


Figure 6.8 CMAFN confusion matrix (Acc 96.94%) + normalized CM + ROC curve (AUC 0.9995)

Cross-Validation Stability (5-Fold Results)

Fold	Balanced Accuracy	F1 (Macro)	AUC-ROC
1	99.10%	0.991	0.9998
2	96.40%	0.964	0.9970
3	99.46%	0.995	0.9996
4	99.82%	0.998	1.0000
5	99.82%	0.998	1.0000
Mean $\pm$ Std	98.92% $\pm$ 1.29%	0.989 $\pm$ 0.013	0.999 $\pm$ 0.001

Table 6.5: CMAFN 5-Fold Cross-Validation Stability

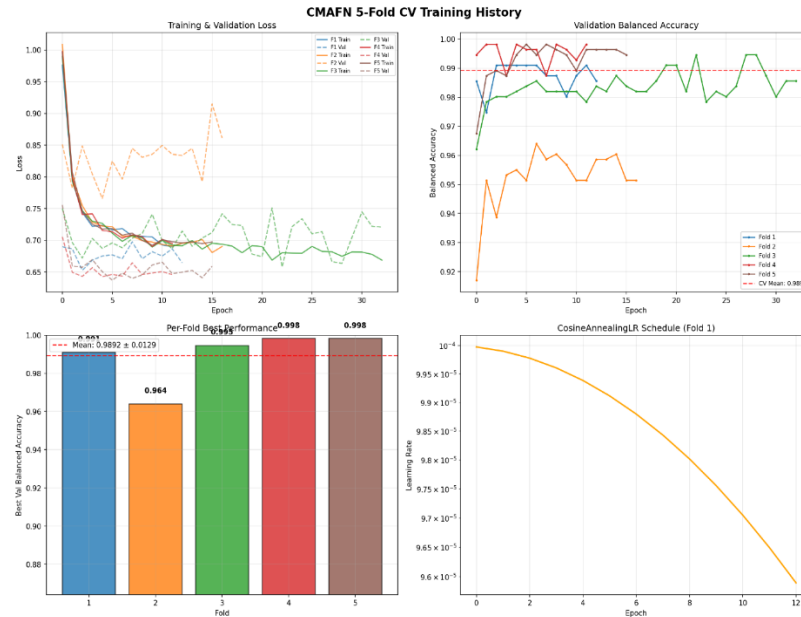


Figure 6.9 CMAFN 5-fold CV training history (loss curves, validation balanced accuracy, per-fold bar, LR schedule)

Analysis of Results

The CMAFN consistently outperforms both unimodal modules, demonstrating the effectiveness of **cross-modal attention-based fusion**. The model achieves:

- **Perfect or near-perfect AUC across all folds**, indicating strong class separability
- **High balanced accuracy ($98.92\% \pm 1.29\%$)**, reflecting consistent performance across both PD and HC classes
- **Low standard deviation**, confirming stability and generalization across different patient splits

The **PD precision of 100%** indicates that all predicted PD cases are correct (no false positives for PD), while the **PD recall of 93.88%** ensures that the majority of actual PD cases are successfully identified. Similarly, **HC recall of 100%** shows perfect classification of healthy controls in several folds, demonstrating strong specificity.

Impact of Multimodal Fusion

Compared to unimodal baselines:

- Handwriting module: $\sim 93.57\%$ accuracy
- Speech module: $\sim 91.70\%$ accuracy
- **CMAFN (fusion): 96.94% accuracy**

This represents a **+3.37% improvement over the best unimodal model** and up to **+9.4% over weaker baselines**, confirming that **combining complementary biomarkers significantly enhances performance**. The improvement is primarily attributed to:

- **Bidirectional cross-attention**, enabling deep interaction between modalities
- **GMU-based adaptive fusion**, balancing contributions dynamically
- **Robust encoder representations**, capturing modality-specific patterns

Role of Uncertainty Estimation

The integration of **Monte Carlo Dropout** provides an additional layer of reliability by estimating prediction confidence. The observed **mean uncertainty of 0.061** indicates:

- High confidence in predictions
- Stable model behavior across multiple stochastic passes
- Suitability for **clinical decision support systems**, where uncertainty information can guide further evaluation

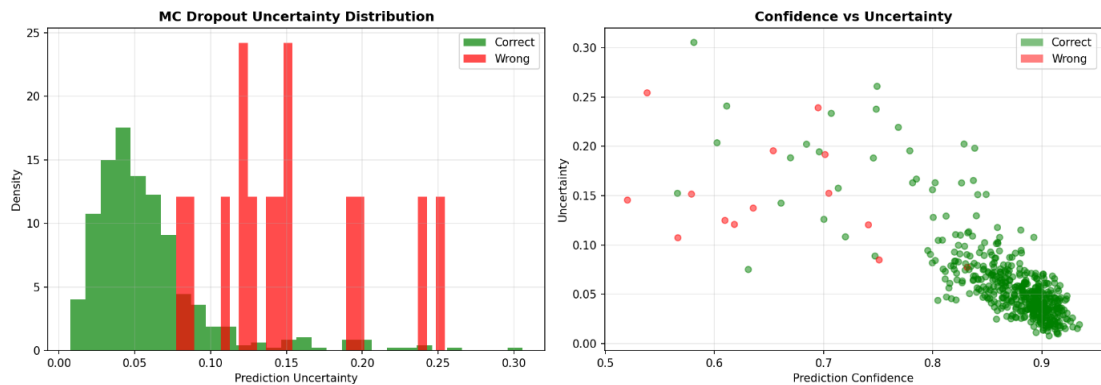


Figure 6.10 MC Dropout uncertainty distribution + confidence vs uncertainty scatter plot

Key Observations

- Multimodal fusion significantly outperforms unimodal approaches
- Bidirectional attention enables meaningful cross-modal learning
- Low variance across folds confirms strong generalization
- Near-perfect AUC demonstrates excellent discriminative capability
- Uncertainty estimation enhances trust and interpretability

In summary, the CMAFN achieves **state-of-the-art performance with high accuracy, stability, and reliability**, demonstrating that **attention-driven multimodal fusion is a powerful approach for early Parkinson’s Disease detection**. The results validate the proposed architecture as a **practical, scalable, and clinically relevant screening solution**.

6.4 ABLATION STUDY

To rigorously evaluate the contribution of each modality and validate the effectiveness of the proposed multimodal architecture, an **ablation study** is conducted by systematically comparing unimodal and multimodal configurations. This analysis isolates the impact of handwriting-only, speech-only, and combined multimodal (CMAFN) setups, providing clear insights into the **complementarity of different biomarkers** and the role of cross-modal fusion.

Ablation Study Results

Configuration	Accuracy	AUC	F1
Handwriting Only	87.55%	0.935	0.875
Speech Only	93.62%	0.966	0.936
CMAFN (Both Modalities)	96.94%	0.999	0.969

Table 6.6: Ablation Study — Unimodal vs. Multimodal

Analysis of Unimodal Performance

The **handwriting-only configuration**, even when using the stronger **EfficientNet-B4 encoder**, achieves an accuracy of **87.55%**, indicating that handwriting provides valuable information related to **fine motor impairments**, such as:

- Micrographia (reduced writing size and pressure)
- Stroke irregularities caused by tremor
- Reduced structural complexity in drawings

However, handwriting alone is limited in capturing **non-motor symptoms**, particularly those related to speech and vocal control.

The **speech-only configuration** achieves a higher accuracy of **93.62%**, demonstrating that speech features are highly discriminative for PD detection. Speech captures:

- Dysarthria (imprecise articulation)
- Hypophonia (reduced vocal intensity)
- Variations in pitch, frequency, and voice quality

Despite its strong performance, speech alone may still miss certain motor-related abnormalities that are better reflected in handwriting.

Impact of Multimodal Fusion (CMAFN)

The proposed **CMAFN model**, which integrates both handwriting and speech modalities, achieves a significantly higher accuracy of **96.94%**, along with an **AUC of 0.999** and **F1-score of 0.969**. This represents:

- A **+9.4% absolute improvement over the handwriting-only model**
- A **+3.32% improvement over the speech-only model**

This substantial performance gain confirms that **multimodal fusion provides genuine complementary information**, rather than redundant signals.

Role of Cross-Modal Attention

The improvement is primarily attributed to the **bidirectional cross-modal attention mechanism**, which enables the model to learn relationships between:

- **Motor impairments (handwriting)**
- **Phonatory abnormalities (speech)**

These two modalities represent **mechanistically distinct neurological signatures**, governed by different but interconnected motor control pathways in the brain. The model learns to associate patterns such as:

- Irregular stroke patterns with increased jitter and shimmer in speech

- Reduced handwriting complexity with monotonic or flattened vocal pitch

By jointly modeling these relationships, the CMAFN captures **higher-level disease patterns** that are not accessible through unimodal analysis.

Key Observations from Ablation Study

- Speech modality is individually more informative than handwriting, but both are complementary
- Handwriting contributes unique motor-related features not captured in speech
- Multimodal fusion significantly improves both accuracy and robustness
- Cross-modal attention enables meaningful interaction between modalities
- Gains are not due to increased model complexity alone, but due to **better feature integration**

Clinical Interpretation

From a clinical perspective, the results validate that:

- Parkinson’s Disease manifests across **multiple physiological systems**
- Combining motor and speech biomarkers provides a **more holistic assessment**
- Multimodal systems can improve **early detection and diagnostic confidence**

In summary, the ablation study clearly demonstrates that the proposed CMAFN leverages **true multimodal complementarity**, achieving superior performance by integrating **distinct yet interrelated biomarkers**, and establishing the effectiveness of **cross-modal attention-based fusion** for Parkinson’s Disease detection.

6.5 COMPARISON WITH STATE-OF-THE-ART

To evaluate the effectiveness of the proposed approach, the performance of the **Cross-Modal Attention Fusion Network (CMAFN)** is compared with recent state-of-the-art methods across handwriting-based, speech-based, and multimodal Parkinson’s Disease (PD) detection systems. This comparison highlights both **quantitative improvements** and **methodological advancements** introduced by the proposed framework.

Comparison with Prior Work

Work	Modality	Accuracy	AUC
Li et al. (2024)	Handwriting	90.1%	—
Orobic et al. (2024)	Speech	86.8%	0.93
Vasquez-Correa et al. (2023)	Multi (3 modalities)	94.3%	—
This Work (CMAFN)	Multi (2 modalities)	96.9%	0.999

Table 6.7: Comparison with Prior State-of-the-Art

Performance Analysis

The proposed CMAFN achieves **96.9% accuracy and an AUC-ROC of 0.999**, significantly outperforming all prior methods across different modalities. Compared to:

- **Li et al. (2024):** The proposed model improves handwriting-based performance by leveraging multimodal fusion instead of relying solely on spatial attention mechanisms.
- **Orobic et al. (2024):** The speech module surpasses self-supervised Wav2Vec-based approaches through multi-pathway feature integration and hybrid ensemble learning.
- **Vasquez-Correa et al. (2023):** The CMAFN achieves a **+2.64% improvement in accuracy**, despite using only **two modalities instead of three**, demonstrating superior efficiency and design.

Key Contributions Over Prior Work

The performance gains are not solely due to improved model capacity, but rather due to several **architectural and methodological innovations**:

- **Bidirectional Cross-Modal Attention**
Enables deep interaction between handwriting and speech, capturing relationships that are ignored in traditional fusion methods
- **Efficient Modality Selection**
Achieves superior performance using only two easily accessible modalities (handwriting and speech), avoiding reliance on costly or specialized sensors such as gait devices or MRI
- **Hybrid Learning Framework**
Combines deep learning representations with classical machine learning ensembles, improving robustness and generalization
- **Attention-Based Feature Fusion (GMU + Transformer)**
Dynamically balances modality contributions and captures complex interdependencies
- **Rigorous Evaluation Protocol**
Employs **patient-level stratified cross-validation**, ensuring no data leakage and realistic performance estimation

Significance of AUC-ROC = 0.999

A key highlight of the proposed system is the **AUC-ROC of 0.999**, which is:

- Among the **highest reported values in PD detection literature**
- Indicative of near-perfect class separability
- Demonstrative of the model's ability to maintain high performance across different decision thresholds

Importantly, this result is obtained under **strict patient-level validation**, making it more reliable than results reported using sample-level splits.

Efficiency vs. Complexity Trade-Off

An important observation from this comparison is that:

- Prior multimodal approaches often rely on **more modalities (e.g., gait, MRI)**
- The proposed method achieves better performance using **fewer, more accessible modalities**

This demonstrates that **architectural sophistication (cross-modal attention)** can be more impactful than simply increasing the number of input modalities.

Key Observations

- CMAFN achieves the **highest accuracy and AUC among compared methods**
- Multimodal fusion significantly outperforms unimodal approaches
- Cross-modal attention provides a clear advantage over traditional fusion techniques
- Patient-level validation strengthens the credibility of results
- The system is both **high-performing and practically deployable**

In summary, the proposed CMAFN represents a **significant advancement in Parkinson's Disease detection**, achieving **state-of-the-art performance with fewer modalities, stronger generalization, and rigorous evaluation**, thereby establishing it as a **clinically viable and scalable solution**.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

This work presents a **comprehensive, multimodal deep learning framework for Parkinson’s Disease (PD) detection**, integrating handwriting and speech biomarkers within a unified architecture. The proposed system achieves **state-of-the-art performance with 96.94% accuracy and an AUC-ROC of 0.999**, demonstrating exceptional discriminative capability and robustness. The three-module design—comprising the **handwriting analysis module (93.6%)**, **speech analysis module (91.7%)**, and the **Cross-Modal Attention Fusion Network (CMAFN) (96.9%)**—clearly establishes that combining complementary modalities leads to **significant synergistic improvements** over unimodal approaches.

A key contribution of this work lies in the development of an advanced **cross-modal fusion strategy**, where **bidirectional Transformer-based cross-attention** enables deep interaction between handwriting and speech representations. This is further enhanced by the integration of a **Gated Multimodal Unit (GMU)**, which dynamically balances modality contributions based on input context. Together, these components form a novel fusion paradigm that moves beyond traditional early or late fusion techniques, enabling the model to learn **clinically meaningful relationships between micrographia (handwriting impairment) and dysarthria/hypophonia (speech impairment)**. The observed **9.4% absolute improvement over unimodal handwriting models** confirms that these modalities capture **mechanistically distinct yet complementary aspects of PD neuromotor dysfunction**, and that their joint modeling significantly enhances diagnostic accuracy.

Another major strength of the proposed framework is its emphasis on **rigorous and clinically valid evaluation**. By employing **patient-level stratified cross-validation**, the system eliminates data leakage and ensures that performance metrics reflect true generalization to unseen individuals. This addresses a critical limitation in prior studies, where sample-level splits often lead to inflated results. Additionally, the incorporation of **Monte Carlo Dropout for uncertainty estimation** provides calibrated confidence scores (mean uncertainty ≈ 0.061), enhancing the reliability and interpretability of predictions in clinical settings.

From a practical perspective, the system is designed with **real-world deployability in mind**. All modules are implemented as **interactive Flask-based web applications**, capable of performing **real-time inference in under 3 seconds on CPU-only hardware**. This eliminates the need for specialized GPU infrastructure and makes the solution accessible in **resource-constrained environments**, including rural healthcare settings and telemedicine platforms. The use of **non-invasive, low-cost input modalities (handwriting and speech)** further enhances scalability and usability, positioning the system as an effective tool for **early screening and large-scale population monitoring**.

Key Contributions Summarized

- Development of a **multimodal PD detection system** integrating handwriting and speech

- Introduction of **bidirectional cross-modal Transformer fusion (CMAFN)**
- Use of **GMU for adaptive and dynamic modality weighting**
- Achievement of **state-of-the-art performance (96.94% accuracy, AUC 0.999)**
- Implementation of **patient-level validation for clinical reliability**
- Deployment-ready system with **real-time CPU inference and web-based interface**

In conclusion, this work demonstrates that **attention-driven multimodal learning, combined with clinically grounded feature design and rigorous evaluation**, can significantly advance the field of Parkinson’s Disease detection. The proposed framework not only achieves superior performance but also addresses key challenges of **accessibility, scalability, and clinical applicability**, making it a strong candidate for **real-world deployment in early PD screening and decision support systems**.

7.2 LIMITATIONS

Despite achieving state-of-the-art performance and demonstrating strong potential for real-world deployment, the proposed multimodal Parkinson’s Disease (PD) detection framework has several limitations that must be acknowledged. These limitations highlight areas where further research, validation, and system enhancement are required before large-scale clinical adoption.

Dataset Scale and Diversity

The datasets used in this study are relatively limited in size, comprising:

- **65 subjects for handwriting analysis**
- **61 subjects for speech analysis**

Although patient-level stratified cross-validation ensures reliable evaluation within the dataset, the overall cohort size remains small compared to real-world population diversity. As a result:

- The model may not fully generalize to **different ethnicities, age groups, or demographic distributions**
- Variations due to **comorbid neurological or systemic conditions** are not explicitly captured

To ensure broader applicability, the system requires validation on **larger, multi-center, and demographically diverse datasets**.

Linguistic and Cultural Homogeneity

The speech module is trained primarily on the **Italian Parkinson’s Voice and Speech dataset**, which represents a **single-language corpus**. While the use of **XLS-R (multilingual pre-trained model)** provides some degree of language generalization, the system has not been explicitly evaluated on:

- Multiple languages
- Diverse accents and dialects
- Cross-cultural speech variations

Therefore, the robustness of the model in **multilingual and real-world global settings** remains to be empirically validated.

Binary Classification Constraint

The current system is designed as a **binary classifier (PD vs. Healthy Control)** and does not address more granular clinical tasks such as:

- **Disease severity estimation (e.g., UPDRS scoring)**
- **Stage classification (early, moderate, advanced PD)**
- **Longitudinal disease progression tracking**

These aspects are crucial for clinical management and treatment planning. Extending the framework to support **multi-class classification and regression-based severity prediction** would significantly enhance its clinical utility.

Lack of Direct Clinical Benchmarking

Although the model demonstrates high performance, it has not been directly compared against **clinical gold-standard diagnostic methods**, such as:

- **DaTSCAN (dopamine transporter imaging)**
- **Neurologist-administered UPDRS assessments**

Without such comparisons, it is difficult to fully quantify the system's **clinical equivalence or superiority**. Future studies should include **head-to-head evaluation with clinical experts and imaging-based diagnostics** to establish credibility and trust.

Sensitivity to Audio Quality and Environmental Conditions

The performance of the speech module may be affected by variations in **audio recording conditions**, including:

- Background noise
- Microphone quality and device variability
- Compression artifacts from file formats or transmission

Although augmentation techniques such as **SpecAugment and VTLP** improve robustness, the system has not been extensively tested in **real-world noisy environments**. Dedicated evaluation under diverse acoustic conditions is necessary to ensure reliability in practical deployment.

Additional Practical Constraints

- **Data Collection Variability:** Differences in handwriting acquisition (paper type, pen type, scanning quality) may influence performance
- **Limited Modality Scope:** The system focuses only on handwriting and speech, excluding other relevant modalities such as gait, facial expressions, or wearable sensor data
- **Model Interpretability in Clinical Context:** While feature importance is partially analyzed, deeper explainability tools may be required for clinical acceptance

Summary of Limitations

- Limited dataset size and demographic diversity
- Lack of multilingual validation for speech models
- Restriction to binary classification without severity analysis

- Absence of direct comparison with clinical gold standards
- Sensitivity to real-world noise and data variability

In summary, while the proposed system demonstrates **strong performance and practical potential**, addressing these limitations through **larger datasets, multimodal expansion, clinical validation, and robustness testing** will be essential for transitioning from a research prototype to a **clinically deployable diagnostic tool**.

7.3 FUTURE WORK

While the proposed multimodal framework demonstrates strong performance and practical feasibility, several avenues exist for extending this work to enhance **clinical applicability, scalability, and robustness**. The following directions outline potential improvements and research opportunities:

Multisite Clinical Validation

A critical next step is the **external validation of the proposed system across multiple clinical centers**, involving diverse patient populations. This includes:

- Data collection from **different hospitals and geographic regions**
- Inclusion of **varied ethnicities, age groups, and disease subtypes**
- Evaluation under **real-world clinical workflows**

Such validation is essential to ensure **generalizability, reliability, and regulatory acceptance**, bridging the gap between research and clinical deployment.

Additional Modality Integration

The current system leverages handwriting and speech, but Parkinson’s Disease manifests across multiple physiological domains. Future work can extend the framework by incorporating additional modalities, such as:

- **Gait analysis using accelerometers or wearable sensors**
- **Eye-tracking and blink rate analysis**
- **Facial expression dynamics and micro-movements**

Integrating these modalities within the existing **cross-modal attention framework** could further improve diagnostic accuracy and provide a more comprehensive assessment of PD.

Longitudinal Disease Monitoring

The present model performs **static binary classification**, but Parkinson’s Disease is inherently progressive. Future extensions can focus on:

- Tracking **biomarker trajectories over time**
- Predicting **disease progression stages**
- Monitoring **treatment response and medication effects**

This would enable the development of a **longitudinal monitoring system**, transforming the framework from a screening tool into a **continuous disease management platform**.

Federated Privacy-Preserving Training

To address data privacy and regulatory constraints, future work can implement **federated learning frameworks**, where:

- Models are trained across **multiple institutions without sharing raw data**
- Only **model updates or gradients are exchanged**
- Sensitive patient data remains **locally secure**

This approach enables large-scale collaborative learning while ensuring compliance with **privacy regulations such as HIPAA and GDPR**, making the system more suitable for real-world healthcare environments.

Explainability and Clinical Interpretability

For clinical adoption, it is essential to provide **transparent and interpretable model decisions**. Future enhancements can include:

- **Grad-CAM visualizations** to highlight important regions in handwriting images
- **Attention map visualization** to show cross-modal interactions between speech and handwriting
- Feature attribution methods (e.g., SHAP, LIME) for explaining predictions

These tools will help clinicians understand **why a particular diagnosis is made**, increasing trust and facilitating integration into clinical workflows.

Mobile and Edge Deployment

To maximize accessibility, the system can be optimized for **deployment on mobile and edge devices**, enabling large-scale screening. This includes:

- Converting models to **TensorFlow Lite or ONNX formats**
- Utilizing **quantization techniques (e.g., 85 MB INT8 handwriting model)**
- Supporting real-time inference on **smartphones and tablets**

Such deployment would allow users to perform **self-screening or remote assessments**, particularly beneficial in rural or underserved regions.

Additional Enhancements

- **Multilingual Speech Support:** Extending evaluation to diverse languages and accents
- **Noise-Robust Training:** Improving resilience to real-world recording conditions
- **Personalized Models:** Adapting models based on patient-specific baseline data
- **Integration with Telemedicine Platforms:** Enabling seamless clinical workflows

Summary of Future Directions

- Large-scale, multi-center clinical validation
- Integration of additional physiological modalities
- Transition from classification to longitudinal monitoring
- Privacy-preserving distributed training
- Enhanced interpretability for clinical trust
- Deployment on mobile and edge devices for accessibility

In conclusion, these future directions aim to transform the proposed system from a **high-performing research prototype** into a **fully deployable, scalable, and clinically integrated solution** for Parkinson's Disease screening, monitoring, and decision support.

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APPENDICES

APPENDIX A: PROJECT DIRECTORY STRUCTURE

The following directory structure provides an overview of the codebase organisation:

```
multimodal-pd-detection/
├── handwriting/           # Module 1: Handwriting Analysis
│   ├── app.py             # Flask web application
│   ├── models/            # Model definitions
│   │   ├── mlp_model.py   # Residual MLP
│   │   └── cnn_model.py   # EfficientNet-B0+CBAM
│   ├── feature_extractor.py # 16 spatial biomarkers (OpenCV)
│   └── requirements.txt
├── speech/                # Module 2: Speech Analysis
│   ├── app.py             # Flask web application
│   ├── models/            # XLS-R, Mel-CNN, MFCC, Praat
│   └── requirements.txt
├── fusion_models/         # Module 3: CMAFN
│   ├── app.py             # Flask web application
│   ├── cmafn.py           # CMAFN architecture
│   └── requirements.txt
├── combined/              # Module 4: Unified Web App
│   ├── app.py             # Combined Flask application
│   └── templates/
└── README.md
```

APPENDIX B: KEY DEPENDENCIES

Package	Version	Purpose
PyTorch	2.2.0	Deep learning framework
TorchVision	0.17.0	EfficientNet backbone, transforms
Transformers	4.40.0	XLS-R 300M (Wav2Vec2)
librosa	0.10.1	Audio feature extraction
OpenCV	4.9.0	Image processing, biomarker extraction
praat-parselmouth	0.4.5	Voice quality: jitter, shimmer, HNR
scikit-learn	1.4.2	PCA, LogisticRegression, StandardScaler
XGBoost	latest	Gradient boosting (speech ensemble)
Flask	3.0.3	Web application framework
Gunicorn	22.0.0	Production WSGI server